Score-based generative modeling through SDEs, PDEs and Optimal controls

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Mathematics of Machine Learning seminar

Contents

1 Score Matching and Denoising Autoencoders

2 Score-based Generative Modeling with SDEs (SGMS)
- Anderson's result on reverse time stochastic processes

3 Diffusion based generative modeling - Optimal control perspective

References

SCORE-BASED GENERATIVE MODELING THROUGH STOCHASTIC DIFFERENTIAL EQUATIONS

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An optimal control perspective on diffusion-based generative modeling

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Denoising Autoencoders (DAE) - single sigmoidal hidden layer

A Connection Between Score Matching and Denoising Autoencoders

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- A training input $\mathbf{x} \in D_n$ is first corrupted by additive gaussian noise of covariance $\sigma^2\mathbf{I}$, yielding corrupted input $\tilde{\mathbf{x}} = \mathbf{x} + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2\mathbf{I})$. This corresponds to conditional density $q_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) = \frac{1}{(2\pi)^{d/2}\sigma^d} e^{-\frac{1}{2\sigma^2}\|\tilde{\mathbf{x}}-\mathbf{x}\|^2}$.
- The corrupted version $\tilde{\mathbf{x}}$ is encoded into a hidden representation $\mathbf{h} \in \mathbb{R}^{d_h}$ through an affine mapping followed by a nonlinearity: $\mathbf{h} = \text{encode}(\tilde{\mathbf{x}}) = \text{sigmoid}(\mathbf{W}\tilde{\mathbf{x}} + \mathbf{b})$, where $\tilde{\mathbf{x}} \in \mathbb{R}^d$, $\mathbf{h} \in (0,1)^{d_h}$, $\mathbf{W}$ is a $d_h \times d$ matrix and $\mathbf{b} \in \mathbb{R}^{d_h}$.
- The hidden representation $\mathbf{h}$ is decoded into reconstruction $\mathbf{x}^r$ through affine mapping: $\mathbf{x}^r = \text{decode}(\mathbf{h}) = \mathbf{W}^T\mathbf{h} + \mathbf{c}$, where $\mathbf{c} \in \mathbb{R}^d$.
- The parameters $\theta = \{\mathbf{W}, \mathbf{b}, \mathbf{c}\}$ are optimized so that the expected squared reconstruction error $\|\mathbf{x}^r - \mathbf{x}\|^2$ is minimized, that is, the objective function being minimized by such a DAE is

$$
\begin{aligned}
J_{DAE\sigma}(\theta) &= \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}},\mathbf{x})}[\|\text{decode}(\text{encode}(\tilde{\mathbf{x}})) - \mathbf{x}\|^2] \\
&= \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}},\mathbf{x})}[\|\mathbf{W}^T\text{sigmoid}(\mathbf{W}\tilde{\mathbf{x}} + \mathbf{b}) + \mathbf{c} - \mathbf{x}\|^2]. \qquad (2.1)
\end{aligned}
$$

Score Matching- Hyvärinen (2005)

$$p(\mathbf{x};\theta) = \frac{1}{Z(\theta)}\exp(-E(\mathbf{x};\theta)).$$

E is called the energy function. Following Hyvärinen (2005), we will call score the gradient of the log density with respect to the data vector: $\psi(\mathbf{x};\theta) = \frac{\partial \log p(\mathbf{x};\theta)}{\partial \mathbf{x}}$. Beware that this use differs slightly from traditional statistics terminology where score usually refers to the derivative of the log likelihood with respect to parameters, whereas here we are talking about a score with respect to the data. The core principle of SM (Hyvärinen, 2005) is to learn θ so that $\psi(\mathbf{x};\theta) = \frac{\partial \log p(\mathbf{x};\theta)}{\partial \mathbf{x}}$ best matches the corresponding score

of the true distribution: $\frac{\partial \log q(\mathbf{x})}{\partial x}$. The corresponding objective function to be minimized is the expected squared error between these two vectors:

$$J_{ESMq}(\theta) = \mathbb{E}_{q(\mathbf{x})}\left[\frac{1}{2}\left\|\psi(\mathbf{x};\theta) - \frac{\partial \log q(\mathbf{x})}{\partial \mathbf{x}}\right\|^2\right].$$

$$\underbrace{\mathbb{E}_{q(\mathbf{x})}\left[\frac{1}{2}\left\|\psi(\mathbf{x};\theta) - \frac{\partial \log q(\mathbf{x})}{\partial \mathbf{x}}\right\|^2\right]}_{J_{ESMq}(\theta)}$$

$$= \underbrace{\mathbb{E}_{q(\mathbf{x})}\left[\frac{1}{2}\left\|\psi(\mathbf{x};\theta)\right\|^2 + \sum_{i=1}^{d}\frac{\partial \psi_i(\mathbf{x};\theta)}{\partial \mathbf{x}_i}\right]}_{J_{ISMq}(\theta)} + C_1, \qquad (3.1)$$

Linking Score Matching to DAE

$$J_{ISMq_0}(\theta) = \mathbb{E}_{q_0(\mathbf{x})}\left[\frac{1}{2}\left\|\psi(\mathbf{x};\theta)\right\|^2 + \sum_{i=1}^{d}\frac{\partial\psi_i(\mathbf{x};\theta)}{\partial\mathbf{x}_i}\right]$$
$$= \frac{1}{n}\sum_{t=1}^{n}\left(\frac{1}{2}\left\|\psi(\mathbf{x}^{(t)};\theta)\right\|^2 + \sum_{i=1}^{d}\frac{\partial\psi_i(\mathbf{x}^{(t)};\theta)}{\partial\mathbf{x}_i}\right). \quad (3.3)$$

$$J_{ESMq_\sigma}(\theta) = \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}})}\left[\frac{1}{2}\left\|\psi(\tilde{\mathbf{x}};\theta) - \frac{\partial\log q_\sigma(\tilde{\mathbf{x}})}{\partial\tilde{\mathbf{x}}}\right\|^2\right]. \quad (4.1)$$

$$J_{DSMq_\sigma}(\theta) = \mathbb{E}_{q_\sigma(\mathbf{x},\tilde{\mathbf{x}})}\left[\frac{1}{2}\left\|\psi(\tilde{\mathbf{x}};\theta) - \frac{\partial\log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})}{\partial\tilde{\mathbf{x}}}\right\|^2\right]. \quad (4.3)$$

$$\frac{\partial\log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})}{\partial\tilde{\mathbf{x}}} = \frac{1}{\sigma^2}(\mathbf{x} - \tilde{\mathbf{x}}). \quad (4.4)$$

$$J_{ESMq_\sigma} \smile J_{DSMq_\sigma}. \quad (4.5)$$

Denoising Score Matching (cont.)

$$p(\mathbf{x};\theta) = \frac{1}{Z(\theta)} \exp(-E(x;\theta)),$$

$$E(\mathbf{x}; \underbrace{\mathbf{W}, \mathbf{b}, \mathbf{c}}_{\theta}) = -\frac{\langle \mathbf{c}, \mathbf{x} \rangle - \frac{1}{2}\|\mathbf{x}\|^2 + \sum_{j=1}^{d_h} \text{softplus}\left(\langle \mathbf{W}_j, \mathbf{x} \rangle + \mathbf{b}_j\right)}{\sigma^2}. \quad (4.6)$$

We then have

$$\begin{aligned}
\psi_i(\mathbf{x};\theta) &= \frac{\partial \log p(\mathbf{x};\theta)}{\partial \mathbf{x}_i} \\
&= -\frac{\partial E}{\partial \mathbf{x}_i} \\
&= \frac{1}{\sigma^2}\left(\mathbf{c}_i - \mathbf{x}_i + \sum_{j=1}^{d_h} \text{softplus}'\left(\langle \mathbf{W}_j, \mathbf{x} \rangle + \mathbf{b}_j\right) \frac{\partial\left(\langle \mathbf{W}_j, \mathbf{x} \rangle + \mathbf{b}_j\right)}{\partial \mathbf{x}_i}\right)
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\sigma^2}\left(\mathbf{c}_i - \mathbf{x}_i + \sum_{j=1}^{d_h} \text{sigmoid}\left(\langle \mathbf{W}_j, \mathbf{x} \rangle + \mathbf{b}_j\right) \mathbf{W}_{ji}\right) \\
&= \frac{1}{\sigma^2}\left(\mathbf{c}_i - \mathbf{x}_i + \langle \mathbf{W}_{\cdot i}, \text{sigmoid}(\mathbf{W}\mathbf{x} + \mathbf{b}) \rangle\right),
\end{aligned}$$

Denoising Score Matching (cont.)

$$\psi(\mathbf{x};\theta) = \frac{1}{\sigma^2}\left(\mathbf{W}^T \operatorname{sigmoid}(\mathbf{W}\mathbf{x}+\mathbf{b}) + \mathbf{c} - \mathbf{x}\right). \tag{4.7}$$

$$\begin{aligned}
J_{DSMq_\sigma}(\theta) &= \mathbb{E}_{q_\sigma(\mathbf{x},\tilde{\mathbf{x}})}\left[\frac{1}{2}\left\|\psi(\tilde{\mathbf{x}};\theta) - \frac{\partial \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})}{\partial \tilde{\mathbf{x}}}\right\|^2\right] \\
&= \mathbb{E}_{q_\sigma(\mathbf{x},\tilde{\mathbf{x}})}\left[\frac{1}{2}\left\|\frac{1}{\sigma^2}(\mathbf{W}^T \operatorname{sigmoid}(\mathbf{W}\tilde{\mathbf{x}}+\mathbf{b}) + \mathbf{c} - \tilde{\mathbf{x}})\right.\right. \\
&\qquad \left.\left. - \frac{1}{\sigma^2}(\mathbf{x}-\tilde{\mathbf{x}})\right\|^2\right] \\
&= \frac{1}{2\sigma^4}\mathbb{E}_{q_\sigma(\mathbf{x},\tilde{\mathbf{x}})}\left[\left\|\mathbf{W}^T \operatorname{sigmoid}(\mathbf{W}\tilde{\mathbf{x}}+\mathbf{b}) + \mathbf{c} - \mathbf{x}\right\|^2\right] \\
&= \frac{1}{2\sigma^4} J_{DAE\sigma}(\theta).
\end{aligned}$$

Score Matching (cont.)

$$J_{ISMq_\sigma} \smile J_{ESMq_\sigma} \smile J_{DSMq_\sigma} \smile J_{DAE\sigma}. \tag{5.1}$$

$$J_{ESMq_\sigma}(\theta) = \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}})}\left[\frac{1}{2}\left\|\psi(\tilde{\mathbf{x}};\theta) - \frac{\partial \log q_\sigma(\tilde{\mathbf{x}})}{\partial \tilde{\mathbf{x}}}\right\|^2\right],$$

which we can develop as

$$J_{ESMq_\sigma}(\theta) = \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}})}\left[\frac{1}{2}\left\|\psi(\tilde{\mathbf{x}};\theta)\right\|^2\right] - S(\theta) + C_2, \tag{A.1}$$

where $C_2 = \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}})}\left[\frac{1}{2}\left\|\frac{\partial \log q_\sigma(\tilde{\mathbf{x}})}{\partial \tilde{\mathbf{x}}}\right\|^2\right]$ is a constant that does not depend on θ, and

$$\begin{aligned}
S(\theta) &= \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}})}\left[\left\langle \psi(\tilde{\mathbf{x}};\theta), \frac{\partial \log q_\sigma(\tilde{\mathbf{x}})}{\partial \tilde{\mathbf{x}}}\right\rangle\right] \\
&= \int_{\tilde{\mathbf{x}}} q_\sigma(\tilde{\mathbf{x}}) \left\langle \psi(\tilde{\mathbf{x}};\theta), \frac{\partial \log q_\sigma(\tilde{\mathbf{x}})}{\partial \tilde{\mathbf{x}}}\right\rangle d\tilde{\mathbf{x}} \\
&= \int_{\tilde{\mathbf{x}}} q_\sigma(\tilde{\mathbf{x}}) \left\langle \psi(\tilde{\mathbf{x}};\theta), \frac{\frac{\partial}{\partial \tilde{\mathbf{x}}} q_\sigma(\tilde{\mathbf{x}})}{q_\sigma(\tilde{\mathbf{x}})}\right\rangle d\tilde{\mathbf{x}} \\
&= \int_{\tilde{\mathbf{x}}} \left\langle \psi(\tilde{\mathbf{x}};\theta), \frac{\partial}{\partial \tilde{\mathbf{x}}} q_\sigma(\tilde{\mathbf{x}})\right\rangle d\tilde{\mathbf{x}} \\
&= \int_{\tilde{\mathbf{x}}} \left\langle \psi(\tilde{\mathbf{x}};\theta), \frac{\partial}{\partial \tilde{\mathbf{x}}} \int_{\mathbf{x}} q_0(\mathbf{x}) q_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) d\mathbf{x}\right\rangle d\tilde{\mathbf{x}}
\end{aligned}$$

Score Matching (cont.)

$$
\begin{aligned}
&= \int_{\tilde{\mathbf{x}}} \left\langle \psi(\tilde{\mathbf{x}};\theta), \int_{\mathbf{x}} q_0(\mathbf{x}) \frac{\partial q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})}{\partial \tilde{\mathbf{x}}} d\mathbf{x} \right\rangle d\tilde{\mathbf{x}} \\
&= \int_{\tilde{\mathbf{x}}} \left\langle \psi(\tilde{\mathbf{x}};\theta), \int_{\mathbf{x}} q_0(\mathbf{x}) q_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) \frac{\partial \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})}{\partial \tilde{\mathbf{x}}} d\mathbf{x} \right\rangle d\tilde{\mathbf{x}} \\
&= \int_{\tilde{\mathbf{x}}} \int_{\mathbf{x}} q_0(\mathbf{x}) q_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) \left\langle \psi(\tilde{\mathbf{x}};\theta), \frac{\partial \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})}{\partial \tilde{\mathbf{x}}} \right\rangle d\mathbf{x} d\tilde{\mathbf{x}} \\
&= \int_{\tilde{\mathbf{x}}} \int_{\mathbf{x}} q_\sigma(\tilde{\mathbf{x}}, \mathbf{x}) \left\langle \psi(\tilde{\mathbf{x}};\theta), \frac{\partial \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})}{\partial \tilde{\mathbf{x}}} \right\rangle d\mathbf{x} d\tilde{\mathbf{x}} \\
&= \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}},\mathbf{x})} \left[\left\langle \psi(\tilde{\mathbf{x}};\theta), \frac{\partial \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})}{\partial \tilde{\mathbf{x}}} \right\rangle \right].
\end{aligned}
$$

Substituting this expression for $S(\theta)$ in equation A.1 yields

$$
\begin{aligned}
J_{ESMq_\sigma}(\theta) = \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}})} \left[\frac{1}{2} \left\| \psi(\tilde{\mathbf{x}};\theta) \right\|^2 \right] \\
- \mathbb{E}_{q_\sigma(\mathbf{x},\tilde{\mathbf{x}})} \left[\left\langle \psi(\tilde{\mathbf{x}};\theta), \frac{\partial \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})}{\partial \tilde{\mathbf{x}}} \right\rangle \right] + C_2. \qquad \text{(A.2)}
\end{aligned}
$$

Score Matching (cont.)

$$J_{DSMq_\sigma}(\theta) = \mathbb{E}_{q_\sigma(\mathbf{x},\tilde{\mathbf{x}})}\left[\frac{1}{2}\left\|\psi(\tilde{\mathbf{x}};\theta) - \frac{\partial \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})}{\partial \tilde{\mathbf{x}}}\right\|^2\right],$$

which we can develop as

$$\begin{aligned} J_{DSMq_\sigma}(\theta) =& \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}})}\left[\frac{1}{2}\left\|\psi(\tilde{\mathbf{x}};\theta)\right\|^2\right] \\ & - \mathbb{E}_{q_\sigma(\mathbf{x},\tilde{\mathbf{x}})}\left[\left\langle \psi(\tilde{\mathbf{x}};\theta), \frac{\partial \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})}{\partial \tilde{\mathbf{x}}}\right\rangle\right] + C_3, \end{aligned} \tag{A.3}$$

where $C_3 = \mathbb{E}_{q_\sigma(\mathbf{x},\tilde{\mathbf{x}})}\left[\frac{1}{2}\left\|\frac{\partial \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})}{\partial \tilde{\mathbf{x}}}\right\|^2\right]$ is a constant that does not depend on θ.

Looking at equations A.2 and A.3, we see that $J_{ESMq_\sigma}(\theta) = J_{DSMq_\sigma}(\theta) + C_2 - C_3$. We have thus shown that the two optimization objectives are equivalent.

Noise Conditional Score Network (NCSN)

Generative Modeling by Estimating Gradients of the Data Distribution

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Suppose our dataset consists of i.i.d. samples $\{\mathbf{x}_i \in \mathbb{R}^D\}_{i=1}^N$ from an unknown data distribution $p_{\text{data}}(\mathbf{x})$. We define the *score* of a probability density $p(\mathbf{x})$ to be $\nabla_{\mathbf{x}} \log p(\mathbf{x})$. The *score network* $\mathbf{s}_{\boldsymbol{\theta}} : \mathbb{R}^D \to \mathbb{R}^D$ is a neural network parameterized by $\boldsymbol{\theta}$, which will be trained to approximate the score of $p_{\text{data}}(\mathbf{x})$. The goal of generative modeling is to use the dataset to learn a model for generating new samples from $p_{\text{data}}(\mathbf{x})$. The framework of score-based generative modeling has two ingredients: score matching and Langevin dynamics.

2.1 Score matching for score estimation

Score matching [24] is originally designed for learning non-normalized statistical models based on i.i.d. samples from an unknown data distribution. Following [53], we repurpose it for score estimation. Using score matching, we can directly train a score network $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})$ to estimate $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$ without training a model to estimate $p_{\text{data}}(\mathbf{x})$ first. Different from the typical usage of score matching, we opt not to use the gradient of an energy-based model as the score network to avoid extra computation due to higher-order gradients. The objective minimizes $\frac{1}{2}\mathbb{E}_{p_{\text{data}}}[\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})\|_2^2]$, which can be shown equivalent to the following up to a constant

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\left[\operatorname{tr}(\nabla_{\mathbf{x}}\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})) + \frac{1}{2}\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})\|_2^2\right], \tag{1}$$

where $\nabla_{\mathbf{x}}\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})$ denotes the Jacobian of $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})$. As shown in [53], under some regularity conditions the minimizer of Eq. (3) (denoted as $\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x})$) satisfies $\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}) = \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$ almost surely. In practice, the expectation over $p_{\text{data}}(\mathbf{x})$ in Eq. (1) can be quickly estimated using data samples. However, score matching is not scalable to deep networks and high dimensional data [53] due to the computation of $\operatorname{tr}(\nabla_{\mathbf{x}}\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}))$. Below we discuss two popular methods for large scale score matching.

NCSN (cont.)

Denoising score matching Denoising score matching [61] is a variant of score matching that completely circumvents $\mathrm{tr}(\nabla_{\mathbf{x}}\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}))$. It first perturbs the data point $\mathbf{x}$ with a pre-specified noise distribution $q_\sigma(\tilde{\mathbf{x}} \mid \mathbf{x})$ and then employs score matching to estimate the score of the perturbed data distribution $q_\sigma(\tilde{\mathbf{x}}) \triangleq \int q_\sigma(\tilde{\mathbf{x}} \mid \mathbf{x})p_{\text{data}}(\mathbf{x})\mathrm{d}\mathbf{x}$. The objective was proved equivalent to the following:

$$\frac{1}{2}\mathbb{E}_{q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})p_{\text{data}}(\mathbf{x})}[\|\mathbf{s}_{\boldsymbol{\theta}}(\tilde{\mathbf{x}}) - \nabla_{\tilde{\mathbf{x}}}\log q_\sigma(\tilde{\mathbf{x}} \mid \mathbf{x})\|_2^2]. \tag{2}$$

As shown in [61], the optimal score network (denoted as $\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x})$) that minimizes Eq. (2) satisfies $\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}) = \nabla_{\mathbf{x}}\log q_\sigma(\mathbf{x})$ almost surely. However, $\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}) = \nabla_{\mathbf{x}}\log q_\sigma(\mathbf{x}) \approx \nabla_{\mathbf{x}}\log p_{\text{data}}(\mathbf{x})$ is true only when the noise is small enough such that $q_\sigma(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x})$.

Sliced score matching Sliced score matching [53] uses random projections to approximate $\mathrm{tr}(\nabla_{\mathbf{x}}\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}))$ in score matching. The objective is

$$\mathbb{E}_{p_{\mathbf{v}}}\mathbb{E}_{p_{\text{data}}}\left[\mathbf{v}^\intercal\nabla_{\mathbf{x}}\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})\mathbf{v} + \frac{1}{2}\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})\|_2^2\right], \tag{3}$$

where $p_{\mathbf{v}}$ is a simple distribution of random vectors, *e.g.*, the multivariate standard normal. As shown in [53], the term $\mathbf{v}^\intercal\nabla_{\mathbf{x}}\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})\mathbf{v}$ can be efficiently computed by forward mode auto-differentiation. Unlike denoising score matching which estimates the scores of *perturbed* data, sliced score matching provides score estimation for the original *unperturbed* data distribution, but requires around four times more computations due to the forward mode auto-differentiation.

NCSN (cont.)

2.2 Sampling with Langevin dynamics

Langevin dynamics can produce samples from a probability density $p(\mathbf{x})$ using only the score function $\nabla_{\mathbf{x}} \log p(\mathbf{x})$. Given a fixed step size $\epsilon > 0$, and an initial value $\tilde{\mathbf{x}}_0 \sim \pi(\mathbf{x})$ with π being a prior distribution, the Langevin method recursively computes the following

$$\tilde{\mathbf{x}}_t = \tilde{\mathbf{x}}_{t-1} + \frac{\epsilon}{2}\nabla_{\mathbf{x}} \log p(\tilde{\mathbf{x}}_{t-1}) + \sqrt{\epsilon}\, \mathbf{z}_t, \tag{4}$$

where $\mathbf{z}_t \sim \mathcal{N}(0, I)$. The distribution of $\tilde{\mathbf{x}}_T$ equals $p(\mathbf{x})$ when $\epsilon \to 0$ and $T \to \infty$, in which case $\tilde{\mathbf{x}}_T$ becomes an exact sample from $p(\mathbf{x})$ under some regularity conditions [62]. When $\epsilon > 0$ and $T < \infty$, a Metropolis-Hastings update is needed to correct the error of Eq. (4), but it can often be ignored in practice [9, 12, 39]. In this work, we assume this error is negligible when ϵ is small and T is large.

Note that sampling from Eq. (4) only requires the score function $\nabla_{\mathbf{x}} \log p(\mathbf{x})$. Therefore, in order to obtain samples from $p_{\text{data}}(\mathbf{x})$, we can first train our score network such that $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$ and then approximately obtain samples with Langevin dynamics using $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})$. This is the key idea of our framework of *score-based generative modeling*.

4.1 Noise Conditional Score Networks

Let $\{\sigma_i\}_{i=1}^L$ be a positive geometric sequence that satisfies $\frac{\sigma_1}{\sigma_2} = \cdots = \frac{\sigma_{L-1}}{\sigma_L} > 1$. Let $q_\sigma(\mathbf{x}) \triangleq \int p_{\text{data}}(\mathbf{t})\mathcal{N}(\mathbf{x} \mid \mathbf{t}, \sigma^2 I)\mathrm{d}\mathbf{t}$ denote the perturbed data distribution. We choose the noise levels $\{\sigma_i\}_{i=1}^L$ such that σ_1 is large enough to mitigate the difficulties discussed in Section 3, and σ_L is small enough to minimize the effect on data. We aim to train a conditional score network to jointly estimate the scores of all perturbed data distributions, *i.e.*, $\forall \sigma \in \{\sigma_i\}_{i=1}^L : \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) \approx \nabla_{\mathbf{x}} \log q_\sigma(\mathbf{x})$. Note that $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) \in \mathbb{R}^D$ when $\mathbf{x} \in \mathbb{R}^D$. We call $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma)$ a *Noise Conditional Score Network (NCSN)*.

NCSN (cont.)

4.2 Learning NCSNs via score matching

Both sliced and denoising score matching can train NCSNs. We adopt denoising score matching as it is slightly faster and naturally fits the task of estimating scores of noise-perturbed data distributions. However, we emphasize that empirically sliced score matching can train NCSNs as well as denoising score matching. We choose the noise distribution to be $q_\sigma(\tilde{\mathbf{x}} \mid \mathbf{x}) = \mathcal{N}(\tilde{\mathbf{x}} \mid \mathbf{x}, \sigma^2 I)$; therefore $\nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}} \mid \mathbf{x}) = -(\tilde{\mathbf{x}} - \mathbf{x})/\sigma^2$. For a given σ, the denoising score matching objective (Eq. (2)) is

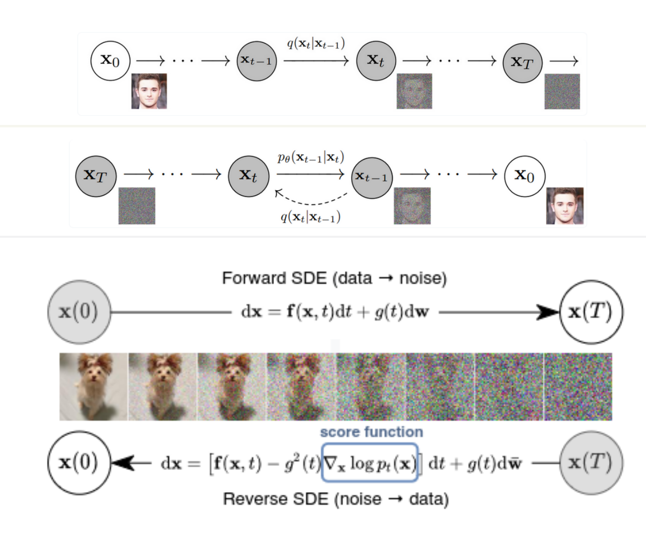
$$\ell(\boldsymbol{\theta}; \sigma) \triangleq \frac{1}{2}\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\tilde{\mathbf{x}} \sim \mathcal{N}(\mathbf{x}, \sigma^2 I)}\left[\left\|\mathbf{s}_{\boldsymbol{\theta}}(\tilde{\mathbf{x}}, \sigma) + \frac{\tilde{\mathbf{x}} - \mathbf{x}}{\sigma^2}\right\|_2^2\right]. \tag{5}$$

Then, we combine Eq. (5) for all $\sigma \in \{\sigma_i\}_{i=1}^L$ to get one unified objective

$$\mathcal{L}(\boldsymbol{\theta}; \{\sigma_i\}_{i=1}^L) \triangleq \frac{1}{L}\sum_{i=1}^{L} \lambda(\sigma_i)\ell(\boldsymbol{\theta}; \sigma_i), \tag{6}$$

where $\lambda(\sigma_i) > 0$ is a coefficient function depending on σ_i. Assuming $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma)$ has enough capacity, $\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}, \sigma)$ minimizes Eq. (6) if and only if $\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}, \sigma_i) = \nabla_{\mathbf{x}} \log q_{\sigma_i}(\mathbf{x})$ a.s. for all $i \in \{1, 2, \cdots, L\}$, because Eq. (6) is a conical combination of L denoising score matching objectives.

Diffusion Models



Anderson's result (1982)

Let $(\Omega, \mathscr{A}, P)$ be a fixed probability space, let $\{\mathscr{A}_t, -\infty < t < \infty\}$ be an increasing family of sub-σ-algebras on $\mathscr{A}$, and let $\{w_t, -\infty < t < \infty\}$ be an r-vector Brownian motion process such that w_t is $\mathscr{A}_t$-measurable for each t, and $w_t - w_s$ for $t \geq s$ is independent of $\mathscr{A}_s$; we require for $s \geq 0$

$$\mathbf{E}[w_{t+s} | \mathscr{A}_t] = w_t, \tag{3.1}$$

$$\mathbf{E}[(w_{t+s} - w_t)(w_{t+s} - w_t)' | \mathscr{A}_t] = sI. \tag{3.2}$$

We study an Ito stochastic differential equation of the form

$$\mathrm{d}x_t = f(x_t, t)\,\mathrm{d}t + g(x_t, t)\,\mathrm{d}w_t. \tag{3.3}$$

We require a decreasing family $\{\bar{\mathscr{A}}_t, -\infty < t < \infty\}$ of sub-σ-algebras on $\mathscr{A}$ and an n-vector process $\{\bar{w}_t, -\infty < t < \infty\}$ such that $\bar{w}_t$ is $\bar{\mathscr{A}}_t$-measurable for each t, $\bar{w}_t - \bar{w}_s$ for $t \geq s$ is independent of $\bar{\mathscr{A}}_t$, and for $s \geq 0$

$$\mathbf{E}[\bar{w}_t | \bar{\mathscr{A}}_{t+s}] = \bar{w}_{t+s}, \tag{3.4}$$

$$\mathbf{E}[(\bar{w}_t - \bar{w}_{t+s})(\bar{w}_t - \bar{w}_{t+s})' | \bar{\mathscr{A}}_{t+s}] = sI. \tag{3.5}$$

This process drives a reverse-time Ito equation of the form

$$\mathrm{d}x_t = \bar{f}(x_t, t)\,\mathrm{d}t + \bar{g}(x_t, t)\,\mathrm{d}\bar{w}_t, \tag{3.6}$$

which is understood to be defined in some region $t \leq T$, where it may be possible to have $T \to \infty$. One has x_T a random variable independent of $\{\bar{w}_t, -\infty < t < \infty\}$ and (3.6) is shorthand for

$$x_T - x_t = \int_t^T \bar{f}(x_t, t)\,\mathrm{d}t + \int_t^T \bar{g}(x_t, t)\,\mathrm{d}\bar{w}_t, \tag{3.7}$$

Anderson's result (cont.)

Theorem. *Let x_t be the process described by (3.3), and suppose $f(\cdot,\cdot)$ and $g(\cdot,\cdot)$ are such as to guarantee the existence of the probability density $p(x_t, t)$ for $t_0 \leqslant t \leqslant T$ as a smooth and unique solution of its associated Kolmogorov equation. Suppose further that an r-vector process $\bar{w}_t$ is defined by $\bar{w}_{t_0} = 0$ and*

$$\mathrm{d}\bar{w}_t^k = \mathrm{d}w_t^k + \frac{1}{p(x_t, t)} \sum_j \frac{\partial}{\partial x_t^j} [p(x_t, t) g^{jk}(x_t, t)]\, \mathrm{d}t, \tag{3.10}$$

and that the forward Kolmogorov equation associated with the joint process $(x_t, \bar{w}_t)$ yields a smooth and unique solution in $t > t_0$ for $p(x_t, \bar{w}_t, t)$ and in $t > s \geqslant t_0$ for $p(x_t, \bar{w}_t, t \mid \bar{w}_s, s)$. Then

(i) *x_t and $\bar{w}_t - \bar{w}_s$ are independent for all $t \geqslant s \geqslant t_0$.*

(ii) *With $\bar{\mathscr{A}}_t$ the minimal σ-algebra with respect to which x_s for $s \geqslant t$ and $\bar{w}_s$ for $s \geqslant t$ are measurable, conditions (3.4) and (3.5) hold.*

(iii) *A reverse time model for x_t is defined by*

$$\mathrm{d}x_t = \bar{f}(x_t, t)\, \mathrm{d}t + g(x_t, t)\, \mathrm{d}\bar{w}_t \tag{3.11}$$

where

$$\bar{f}^i(x_t, t) = f^i(x_t, t) - \frac{1}{p(x_t, t)} \sum_{j,k} \frac{\partial}{\partial x_t^j} [p(x_t, t) g^{ik}(x_t, t) g^{jk}(x_t, t)]. \tag{3.12}$$

Anderson's result (cont.)

Let us consider the joint process $(x_t, \bar{w}_t)$ defined by

$$dx_t = f(x_t, t)\,dt + g(x_t, t)\,dw_t, \tag{4.1}$$

$$d\bar{w}_t^k = \frac{1}{p(x_t, t)} \sum_j \frac{\partial}{\partial x_t^j}[p(x_t, t) g^{jk}(x_t, t)]\,dt + dw_t^k, \tag{4.2}$$

with $k = 1, \ldots, r$. The associated forward Kolmogorov equation is

$$\begin{aligned}
\frac{\partial p(x_t, \bar{w}_t, t)}{\partial t} = &- \sum_{i=1}^{n} \frac{\partial}{\partial x_t^i}[p(x_t, \bar{w}_t, t) f^i(x_t, t)] \\
&- \sum_{k=1}^{r} \frac{\partial}{\partial \bar{w}_t^k}\left\{\frac{p(x_t, \bar{w}_t, t)}{p(x_t, t)} \sum_j \frac{\partial}{\partial x_t^j}[p(x_t, t) g^{jk}(x_t, t)]\right\} \\
&+ \tfrac{1}{2} \sum_{i,j=1}^{n} \frac{\partial^2}{\partial x_t^i\, \partial x_t^j}\{p(x_t, \bar{w}_t, t)[g(x_t, t) g'(x_t, t)]^{ij}\} \\
&+ \tfrac{1}{2} \sum_{k,l=1}^{r} \frac{\partial^2}{\partial \bar{w}_t^k\, \partial \bar{w}_t^l}[p(x_t, \bar{w}_t, t)] \\
&+ \sum_{i=1}^{n} \sum_{k=1}^{r} \frac{\partial^2}{\partial x_t^i\, \partial \bar{w}_t^k}[p(x_t, \bar{w}_t, t) g^{ik}(x_t, t)],
\end{aligned} \tag{4.3}$$

and the boundary condition we take is the natural one

$$p(x_{t_0}, \bar{w}_{t_0}, t_0) = p(x_{t_0}, t_0)\,\delta(\bar{w}_{t_0}). \tag{4.4}$$

Anderson's result (cont.)

Lemma 1. *Suppose $p(x_t, t)$ is the solution of the forward Kolmogorov equation for* (4.1) *and*

$$\phi(\bar{w}_t, t) = \frac{1}{[2\pi(t-t_0)]^{r/2}} \exp\left[-\frac{\bar{w}_t' \bar{w}_t}{2(t-t_0)}\right]. \tag{4.5}$$

Then the solution of (4.3) *and* (4.4) *is given by*

$$p(x_t, \bar{w}_t, t) = p(x_t, t)\phi(\bar{w}_t, t). \tag{4.6}$$

Lemma 2. *With the same hypotheses as Lemma* 1,

$$p(x_t, \bar{w}_t, t) = p(x_t, t)p(\bar{w}_t, t). \tag{4.8}$$

Lemma 3. *Suppose $p(x_t, t)$ is the solution of the forward Kolmogorov equation for* (4.1). *Then the conditional density, $p(x_t, \bar{w}_t, t \mid \bar{w}_s, s)$ associated with* (4.1) *and* (4.2) *for $t \geqslant s \geqslant t_0$ is*

$$p(x_t, \bar{w}_t, t \mid \bar{w}_s, s) = p(x_t, t)\psi(\bar{w}_t, \bar{w}_s, t-s) \tag{4.9}$$

where

$$\psi(\bar{w}_t, \bar{w}_s, t-s) = \frac{1}{[2\pi(t-s)]^{r/2}} \exp\left[-\frac{(\bar{w}_t - \bar{w}_s)'(\bar{w}_t - \bar{w}_s)}{2(t-s)}\right]. \tag{4.10}$$

Anderson's result (cont.)

Lemma 4. *Let A, B, C be three jointly distributed random variables, and let $p_A(a)$, etc. denote the probability density of A evaluated at A = a. Then, if*

$$p_{B|C}(b|c) = f(b-c)$$

for some function f, defining D = B − C results in

$$p_D(d) = f(d).$$

B.D.O. Anderson / Reverse-time diffusion equation 321

If

$$p_{A,B|C}(a, b|c) = p_A(a) f(b-c)$$

for some function f, defining D = B − C results in

$$p_{A,D} = p_A(a) p_D(d) = p_A(a) f(d).$$

Now we have the following lemma.

Lemma 5. *With x_t, $\bar{w}_t$ defined as above, and for $t \geq s \geq t_0$,*

$$p(x_t, \bar{w}_t - \bar{w}_s, t, s) = p(x_t, t) p(\bar{w}_t - \bar{w}_s, t, s).$$

Anderson's result (cont.)

$$dx_t = f(x_t, t)\,dt + g(x_t, t)\,dw_t \tag{5.1}$$

where w_t has the usual properties. Then the backward (not reverse-time) Kolmogorov equation for $s \geq t$ is

$$-\frac{\partial p(x_s, s \mid x_t, t)}{\partial t} = \sum_i f^i(x_t, t)\frac{\partial p(x_s, s \mid x_t, t)}{\partial x_t^i} + \frac{1}{2}\sum_{i,j,k} g^{ik}(x_t, t)g^{jk}(x_t, t)\frac{\partial^2 p(x_s, s \mid x_t, t)}{\partial x_t^i\,\partial x_t^j}, \tag{5.2}$$

and the forward, unconditioned, equation yields

$$-\frac{\partial p(x_t, t)}{\partial t} = \sum_i \frac{\partial}{\partial x_t^i}[p(x_t, t)f^i(x_t, t)] - \frac{1}{2}\sum_{i,j,k}\frac{\partial^2[g^{ik}(x_t, t)g^{jk}(x_t, t)p(x_t, t)]}{\partial x_t^i\,\partial x_t^j}. \tag{5.3}$$

Now, because

$$p(x_t, t, x_s, s) = p(x_s, s \mid x_t, t)\,p(x_t, t), \tag{5.4}$$

we can attempt to obtain a partial differential equation for $p(x_t, t, x_s, s)$, regarding x_t, t as the independent variables and x_s, s as parameters. We obtain, combining (5.2) through (5.4),

$$\frac{\partial p(x_t, t, x_s, s)}{\partial t} = \text{terms involving } f,\ g,\ p(x_t, t) \text{ and } p(x_s, s \mid x_t, t) \text{ and their } x_t\text{-derivatives.}$$

Anderson's result (cont.)

$$-\frac{\partial}{\partial t} p(x_t, t, x_s, s) = \sum_i \frac{\partial}{\partial x_t^i} [\bar{f}^i(x_t, t) p(x_t, t, x_s, s)]$$
$$+\frac{1}{2} \sum_{i,j,k} \frac{\partial^2 [p(x_t, t, x_s, s) g^{ik}(x_t, t) g^{jk}(x_t, t)]}{\partial x_t^i \partial x_t^j} \tag{5.5}$$

where $\bar{f}^i$ is as before, viz,

$$\bar{f}^i(x_t, t) = f^i(x_t, t) - \frac{1}{p(x_t, t)} \sum_{j,k} \frac{\partial}{\partial x_t^j} [p(x_t, t) g^{ik}(x_t, t) g^{jk}(x_t, t)]. \tag{5.6}$$

The same partial differential equation (but with different boundary conditions of course) is satisfied by $p(x_t, t | x_s, s)$ [and in fact $p(x_t, t)$]—this is trivial to see. Just as (5.3) corresponds to the forward model (5.1), so then (5.5) has to correspond to the reverse model

$$\mathrm{d}\bar{x}_t = \bar{f}(\bar{x}_t, t)\, \mathrm{d}t + g(\bar{x}_t, t)\, \mathrm{d}\bar{w}_t \tag{5.7}$$

where $\bar{w}_t$ is a vector Wiener process with past increments independent of $\bar{x}_t$. (The

Denoising Score Matching with Langevin Dynamics (SMLD)

Let $p_\sigma(\tilde{\mathbf{x}} \mid \mathbf{x}) := \mathcal{N}(\tilde{\mathbf{x}}; \mathbf{x}, \sigma^2\mathbf{I})$ be a perturbation kernel, and $p_\sigma(\tilde{\mathbf{x}}) := \int p_{\mathrm{data}}(\mathbf{x}) p_\sigma(\tilde{\mathbf{x}} \mid \mathbf{x}) \mathrm{d}\mathbf{x}$, where $p_{\mathrm{data}}(\mathbf{x})$ denotes the data distribution. Consider a sequence of positive noise scales $\sigma_{\min} = \sigma_1 < \sigma_2 < \cdots < \sigma_N = \sigma_{\max}$. Typically, $\sigma_{\min}$ is small enough such that $p_{\sigma_{\min}}(\mathbf{x}) \approx p_{\mathrm{data}}(\mathbf{x})$, and $\sigma_{\max}$ is

large enough such that $p_{\sigma_{\max}}(\mathbf{x}) \approx \mathcal{N}(\mathbf{x}; \mathbf{0}, \sigma_{\max}^2 \mathbf{I})$. Song & Ermon (2019) propose to train a Noise Conditional Score Network (NCSN), denoted by $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma)$, with a weighted sum of denoising score matching (Vincent, 2011) objectives:

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{N} \sigma_i^2 \mathbb{E}_{p_{\mathrm{data}}(\mathbf{x})} \mathbb{E}_{p_{\sigma_i}(\tilde{\mathbf{x}} \mid \mathbf{x})} \left[\left\| \mathbf{s}_{\boldsymbol{\theta}}(\tilde{\mathbf{x}}, \sigma_i) - \nabla_{\tilde{\mathbf{x}}} \log p_{\sigma_i}(\tilde{\mathbf{x}} \mid \mathbf{x}) \right\|_2^2 \right]. \tag{1}$$

Given sufficient data and model capacity, the optimal score-based model $\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}, \sigma)$ matches $\nabla_{\mathbf{x}} \log p_\sigma(\mathbf{x})$ almost everywhere for $\sigma \in \{\sigma_i\}_{i=1}^N$. For sampling, Song & Ermon (2019) run M steps of Langevin MCMC to get a sample for each $p_{\sigma_i}(\mathbf{x})$ sequentially:

$$\mathbf{x}_i^m = \mathbf{x}_i^{m-1} + \epsilon_i \mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}_i^{m-1}, \sigma_i) + \sqrt{2\epsilon_i}\,\mathbf{z}_i^m, \quad m = 1, 2, \cdots, M, \tag{2}$$

where $\epsilon_i > 0$ is the step size, and $\mathbf{z}_i^m$ is standard normal. The above is repeated for $i = N, N-1, \cdots, 1$ in turn with $\mathbf{x}_N^0 \sim \mathcal{N}(\mathbf{x} \mid \mathbf{0}, \sigma_{\max}^2 \mathbf{I})$ and $\mathbf{x}_i^0 = \mathbf{x}_{i+1}^M$ when $i < N$. As $M \to \infty$ and $\epsilon_i \to 0$ for all i, $\mathbf{x}_1^M$ becomes an exact sample from $p_{\sigma_{\min}}(\mathbf{x}) \approx p_{\mathrm{data}}(\mathbf{x})$ under some regularity conditions.

Denoising Diffusion Probabilistic Models (DDPM)

Sohl-Dickstein et al. (2015); Ho et al. (2020) consider a sequence of positive noise scales $0 < \beta_1, \beta_2, \cdots, \beta_N < 1$. For each training data point $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, a discrete Markov chain $\{\mathbf{x}_0, \mathbf{x}_1, \cdots, \mathbf{x}_N\}$ is constructed such that $p(\mathbf{x}_i \mid \mathbf{x}_{i-1}) = \mathcal{N}(\mathbf{x}_i; \sqrt{1-\beta_i}\mathbf{x}_{i-1}, \beta_i \mathbf{I})$, and therefore $p_{\alpha_i}(\mathbf{x}_i \mid \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_i; \sqrt{\alpha_i}\mathbf{x}_0, (1-\alpha_i)\mathbf{I})$, where $\alpha_i := \prod_{j=1}^{i}(1-\beta_j)$. Similar to SMLD, we can denote the perturbed data distribution as $p_{\alpha_i}(\tilde{\mathbf{x}}) := \int p_{\text{data}}(\mathbf{x}) p_{\alpha_i}(\tilde{\mathbf{x}} \mid \mathbf{x}) \mathrm{d}\mathbf{x}$. The noise scales are prescribed such that $\mathbf{x}_N$ is approximately distributed according to $\mathcal{N}(\mathbf{0}, \mathbf{I})$. A variational Markov chain in the reverse direction is parameterized with $p_{\boldsymbol{\theta}}(\mathbf{x}_{i-1}|\mathbf{x}_i) = \mathcal{N}(\mathbf{x}_{i-1}; \frac{1}{\sqrt{1-\beta_i}}(\mathbf{x}_i + \beta_i \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_i, i)), \beta_i \mathbf{I})$, and trained with a re-weighted variant of the evidence lower bound (ELBO):

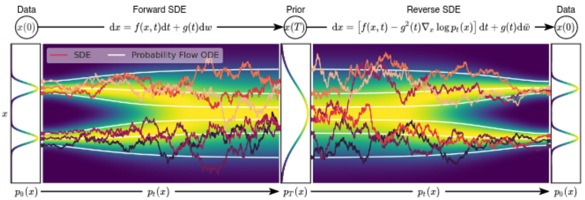
$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{N} (1-\alpha_i) \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{p_{\alpha_i}(\tilde{\mathbf{x}}|\mathbf{x})} [\| \mathbf{s}_{\boldsymbol{\theta}}(\tilde{\mathbf{x}}, i) - \nabla_{\tilde{\mathbf{x}}} \log p_{\alpha_i}(\tilde{\mathbf{x}} \mid \mathbf{x}) \|_2^2]. \tag{3}$$

After solving Eq. (3) to get the optimal model $\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}, i)$, samples can be generated by starting from $\mathbf{x}_N \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and following the estimated reverse Markov chain as below

$$\mathbf{x}_{i-1} = \frac{1}{\sqrt{1-\beta_i}} (\mathbf{x}_i + \beta_i \mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}_i, i)) + \sqrt{\beta_i}\mathbf{z}_i, \quad i = N, N-1, \cdots, 1. \tag{4}$$

We call this method *ancestral sampling*, since it amounts to performing ancestral sampling from the graphical model $\prod_{i=1}^{N} p_{\boldsymbol{\theta}}(\mathbf{x}_{i-1} \mid \mathbf{x}_i)$. The objective Eq. (3) described here is L_{simple} in Ho et al. (2020), written in a form to expose more similarity to Eq. (1). Like Eq. (1), Eq. (3) is also a weighted sum of denoising score matching objectives, which implies that the optimal model, $\mathbf{s}_{\boldsymbol{\theta}^*}(\tilde{\mathbf{x}}, i)$, matches the score of the perturbed data distribution, $\nabla_{\mathbf{x}} \log p_{\alpha_i}(\mathbf{x})$. Notably, the weights of the i-th summand in Eq. (1) and Eq. (3), namely σ_i^2 and $(1-\alpha_i)$, are related to corresponding perturbation kernels in the same functional form: $\sigma_i^2 \propto 1/\mathbb{E}[\|\nabla_{\mathbf{x}} \log p_{\sigma_i}(\tilde{\mathbf{x}} \mid \mathbf{x})\|_2^2]$ and $(1-\alpha_i) \propto 1/\mathbb{E}[\|\nabla_{\mathbf{x}} \log p_{\alpha_i}(\tilde{\mathbf{x}} \mid \mathbf{x})\|_2^2]$.

Generating samples by reverting the SDE



Our goal is to construct a diffusion process $\{\mathbf{x}(t)\}_{t=0}^{T}$ indexed by a continuous time variable $t \in [0, T]$, such that $\mathbf{x}(0) \sim p_0$, for which we have a dataset of i.i.d. samples, and $\mathbf{x}(T) \sim p_T$, for which we have a tractable form to generate samples efficiently. In other words, p_0 is the data distribution and p_T is the prior distribution. This diffusion process can be modeled as the solution to an Itô SDE:

$$\mathrm{d}\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\mathrm{d}t + g(t)\mathrm{d}\mathbf{w}, \tag{5}$$

By starting from samples of $\mathbf{x}(T) \sim p_T$ and reversing the process, we can obtain samples $\mathbf{x}(0) \sim p_0$. A remarkable result from Anderson (1982) states that the reverse of a diffusion process is also a diffusion process, running backwards in time and given by the reverse-time SDE:

$$\mathrm{d}\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x})]\mathrm{d}t + g(t)\mathrm{d}\bar{\mathbf{w}}, \tag{6}$$

where $\bar{\mathbf{w}}$ is a standard Wiener process when time flows backwards from T to 0, and $\mathrm{d}t$ is an infinitesimal negative timestep. Once the score of each marginal distribution, $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$, is known for all t, we can derive the reverse diffusion process from Eq. (6) and simulate it to sample from p_0.

SGMS (cont.)

3.3 Estimating scores for the SDE

The score of a distribution can be estimated by training a score-based model on samples with score matching (Hyvärinen, 2005; Song et al., 2019a). To estimate $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$, we can train a time-dependent score-based model $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, t)$ via a continuous generalization to Eqs. (1) and (3):

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \mathbb{E}_t \Big\{ \lambda(t) \mathbb{E}_{\mathbf{x}(0)} \mathbb{E}_{\mathbf{x}(t)|\mathbf{x}(0)} \big[\left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log p_{0t}(\mathbf{x}(t) \mid \mathbf{x}(0)) \right\|_2^2 \big] \Big\}. \tag{7}$$

Here $\lambda : [0, T] \to \mathbb{R}_{>0}$ is a positive weighting function, t is uniformly sampled over $[0, T]$, $\mathbf{x}(0) \sim p_0(\mathbf{x})$ and $\mathbf{x}(t) \sim p_{0t}(\mathbf{x}(t) \mid \mathbf{x}(0))$. With sufficient data and model capacity, score matching ensures that the optimal solution to Eq. (7), denoted by $\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}, t)$, equals $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ for almost all $\mathbf{x}$ and t. As in SMLD and DDPM, we can typically choose $\lambda \propto 1/\mathbb{E}\big[\left\| \nabla_{\mathbf{x}(t)} \log p_{0t}(\mathbf{x}(t) \mid \mathbf{x}(0)) \right\|_2^2 \big]$. Note that Eq. (7) uses denoising score matching, but other score matching objectives, such as sliced

5 Controllable generation

The continuous structure of our framework allows us to not only produce data samples from p_0, but also from $p_0(\mathbf{x}(0) \mid \mathbf{y})$ if $p_t(\mathbf{y} \mid \mathbf{x}(t))$ is known. Given a forward SDE as in Eq. (5), we can sample

from $p_t(\mathbf{x}(t) \mid \mathbf{y})$ by starting from $p_T(\mathbf{x}(T) \mid \mathbf{y})$ and solving a conditional reverse-time SDE:

$$\mathrm{d}\mathbf{x} = \{\mathbf{f}(\mathbf{x}, t) - g(t)^2[\nabla_{\mathbf{x}} \log p_t(\mathbf{x}) + \nabla_{\mathbf{x}} \log p_t(\mathbf{y} \mid \mathbf{x})]\}\mathrm{d}t + g(t)\mathrm{d}\bar{\mathbf{w}}. \tag{14}$$

In general, we can use Eq. (14) to solve a large family of *inverse problems* with score-based generative models, once given an estimate of the gradient of the forward process, $\nabla_{\mathbf{x}} \log p_t(\mathbf{y} \mid \mathbf{x}(t))$. In some cases, it is possible to train a separate model to learn the forward process $\log p_t(\mathbf{y} \mid \mathbf{x}(t))$ and compute its gradient. Otherwise, we may estimate the gradient with heuristics and domain knowledge.

SGMS (cont.)

A THE FRAMEWORK FOR MORE GENERAL SDEs

In the main text, we introduced our framework based on a simplified SDE Eq. (5) where the diffusion coefficient is independent of $\mathbf{x}(t)$. It turns out that our framework can be extended to hold for more general diffusion coefficients. We can consider SDEs in the following form:

$$\mathrm{d}\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\mathrm{d}t + \mathbf{G}(\mathbf{x}, t)\mathrm{d}\mathbf{w}, \tag{15}$$

where $\mathbf{f}(\cdot, t) : \mathbb{R}^d \to \mathbb{R}^d$ and $\mathbf{G}(\cdot, t) : \mathbb{R}^d \to \mathbb{R}^{d\times d}$. We follow the Itô interpretation of SDEs throughout this paper.

According to (Anderson, 1982), the reverse-time SDE is given by (*cf*., Eq. (6))

$$\mathrm{d}\mathbf{x} = \{\mathbf{f}(\mathbf{x}, t) - \nabla\cdot[\mathbf{G}(\mathbf{x}, t)\mathbf{G}(\mathbf{x}, t)^{\mathsf{T}}] - \mathbf{G}(\mathbf{x}, t)\mathbf{G}(\mathbf{x}, t)^{\mathsf{T}}\nabla_{\mathbf{x}} \log p_t(\mathbf{x})\}\mathrm{d}t + \mathbf{G}(\mathbf{x}, t)\mathrm{d}\bar{\mathbf{w}}, \tag{16}$$

where we define $\nabla\cdot\mathbf{F}(\mathbf{x}) := (\nabla\cdot\mathbf{f}^1(\mathbf{x}), \nabla\cdot\mathbf{f}^2(\mathbf{x}), \cdots, \nabla\cdot\mathbf{f}^d(\mathbf{x}))^{\mathsf{T}}$ for a matrix-valued function $\mathbf{F}(\mathbf{x}) := (\mathbf{f}^1(\mathbf{x}), \mathbf{f}^2(\mathbf{x}), \cdots, \mathbf{f}^d(\mathbf{x}))^{\mathsf{T}}$ throughout the paper.

The probability flow ODE corresponding to Eq. (15) has the following form (*cf*., Eq. (13), see a detailed derivation in Appendix D.1):

$$\mathrm{d}\mathbf{x} = \left\{\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}\nabla\cdot[\mathbf{G}(\mathbf{x}, t)\mathbf{G}(\mathbf{x}, t)^{\mathsf{T}}] - \frac{1}{2}\mathbf{G}(\mathbf{x}, t)\mathbf{G}(\mathbf{x}, t)^{\mathsf{T}}\nabla_{\mathbf{x}} \log p_t(\mathbf{x})\right\}\mathrm{d}t. \tag{17}$$

Finally for conditional generation with the general SDE Eq. (15), we can solve the conditional reverse-time SDE below (*cf*., Eq. (14), details in Appendix I):

$$\begin{aligned}\mathrm{d}\mathbf{x} = \{&\mathbf{f}(\mathbf{x}, t) - \nabla\cdot[\mathbf{G}(\mathbf{x}, t)\mathbf{G}(\mathbf{x}, t)^{\mathsf{T}}] - \mathbf{G}(\mathbf{x}, t)\mathbf{G}(\mathbf{x}, t)^{\mathsf{T}}\nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \\ &- \mathbf{G}(\mathbf{x}, t)\mathbf{G}(\mathbf{x}, t)^{\mathsf{T}}\nabla_{\mathbf{x}} \log p_t(\mathbf{y} \mid \mathbf{x})\}\mathrm{d}t + \mathbf{G}(\mathbf{x}, t)\mathrm{d}\bar{\mathbf{w}}.\end{aligned} \tag{18}$$

When the drift and diffusion coefficient of an SDE are not affine, it can be difficult to compute the transition kernel $p_{0t}(\mathbf{x}(t) \mid \mathbf{x}(0))$ in closed form. This hinders the training of score-based models, because Eq. (7) requires knowing $\nabla_{\mathbf{x}(t)} \log p_{0t}(\mathbf{x}(t) \mid \mathbf{x}(0))$. To overcome this difficulty, we can replace denoising score matching in Eq. (7) with other efficient variants of score matching that do not require computing $\nabla_{\mathbf{x}(t)} \log p_{0t}(\mathbf{x}(t) \mid \mathbf{x}(0))$. For example, when using sliced score matching (Song et al., 2019a), our training objective Eq. (7) becomes

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \mathbb{E}_t\left\{\lambda(t)\mathbb{E}_{\mathbf{x}(0)}\mathbb{E}_{\mathbf{x}(t)}\mathbb{E}_{\mathbf{v}\sim p_{\mathbf{v}}}\left[\frac{1}{2}\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)\|_2^2 + \mathbf{v}^{\mathsf{T}}\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)\mathbf{v}\right]\right\}, \tag{19}$$

where $\lambda : [0, T] \to \mathbb{R}^+$ is a positive weighting function, $t \sim \mathcal{U}(0, T)$, $\mathbb{E}[\mathbf{v}] = \mathbf{0}$, and $\mathrm{Cov}[\mathbf{v}] = \mathbf{I}$. We can always simulate the SDE to sample from $p_{0t}(\mathbf{x}(t) \mid \mathbf{x}(0))$, and solve Eq. (19) to train the time-dependent score-based model $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, t)$.

SGMS (cont.)

consider the SDE in Eq. (15), which possesses the following form:

$$\mathrm{d}\mathbf{x} = \mathbf{f}(\mathbf{x},t)\mathrm{d}t + \mathbf{G}(\mathbf{x},t)\mathrm{d}\mathbf{w},$$

where $\mathbf{f}(\cdot,t) : \mathbb{R}^d \to \mathbb{R}^d$ and $\mathbf{G}(\cdot,t) : \mathbb{R}^d \to \mathbb{R}^{d\times d}$. The marginal probability density $p_t(\mathbf{x}(t))$ evolves according to Kolmogorov's forward equation (Fokker-Planck equation) (Øksendal, 2003)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\sum_{i=1}^d \frac{\partial}{\partial x_i}[f_i(\mathbf{x},t)p_t(\mathbf{x})] + \frac{1}{2}\sum_{i=1}^d\sum_{j=1}^d \frac{\partial^2}{\partial x_i \partial x_j}\Big[\sum_{k=1}^d G_{ik}(\mathbf{x},t)G_{jk}(\mathbf{x},t)p_t(\mathbf{x})\Big]. \tag{35}$$

We can easily rewrite Eq. (35) to obtain

$$\begin{aligned}\frac{\partial p_t(\mathbf{x})}{\partial t} &= -\sum_{i=1}^d \frac{\partial}{\partial x_i}[f_i(\mathbf{x},t)p_t(\mathbf{x})] + \frac{1}{2}\sum_{i=1}^d\sum_{j=1}^d \frac{\partial^2}{\partial x_i \partial x_j}\Big[\sum_{k=1}^d G_{ik}(\mathbf{x},t)G_{jk}(\mathbf{x},t)p_t(\mathbf{x})\Big] \\ &= -\sum_{i=1}^d \frac{\partial}{\partial x_i}[f_i(\mathbf{x},t)p_t(\mathbf{x})] + \frac{1}{2}\sum_{i=1}^d \frac{\partial}{\partial x_i}\Big[\sum_{j=1}^d \frac{\partial}{\partial x_j}\Big[\sum_{k=1}^d G_{ik}(\mathbf{x},t)G_{jk}(\mathbf{x},t)p_t(\mathbf{x})\Big]\Big]. \end{aligned}\tag{36}$$

Note that

$$\begin{aligned}&\sum_{j=1}^d \frac{\partial}{\partial x_j}\Big[\sum_{k=1}^d G_{ik}(\mathbf{x},t)G_{jk}(\mathbf{x},t)p_t(\mathbf{x})\Big] \\ &= \sum_{j=1}^d \frac{\partial}{\partial x_j}\Big[\sum_{k=1}^d G_{ik}(\mathbf{x},t)G_{jk}(\mathbf{x},t)\Big]p_t(\mathbf{x}) + \sum_{j=1}^d\sum_{k=1}^d G_{ik}(\mathbf{x},t)G_{jk}(\mathbf{x},t)p_t(\mathbf{x})\frac{\partial}{\partial x_j}\log p_t(\mathbf{x}) \\ &= p_t(\mathbf{x})\nabla\cdot[\mathbf{G}(\mathbf{x},t)\mathbf{G}(\mathbf{x},t)^{\mathsf{T}}] + p_t(\mathbf{x})\mathbf{G}(\mathbf{x},t)\mathbf{G}(\mathbf{x},t)^{\mathsf{T}}\nabla_{\mathbf{x}}\log p_t(\mathbf{x}),\end{aligned}$$

based on which we can continue the rewriting of Eq. (36) to obtain

SGMS (cont.)

$$
\begin{aligned}
\frac{\partial p_t(\mathbf{x})}{\partial t} &= -\sum_{i=1}^{d} \frac{\partial}{\partial x_i}[f_i(\mathbf{x},t)p_t(\mathbf{x})] + \frac{1}{2}\sum_{i=1}^{d} \frac{\partial}{\partial x_i}\Big[\sum_{j=1}^{d} \frac{\partial}{\partial x_j}\Big[\sum_{k=1}^{d} G_{ik}(\mathbf{x},t)G_{jk}(\mathbf{x},t)p_t(\mathbf{x})\Big]\Big] \\
&= -\sum_{i=1}^{d} \frac{\partial}{\partial x_i}[f_i(\mathbf{x},t)p_t(\mathbf{x})] \\
&\quad + \frac{1}{2}\sum_{i=1}^{d} \frac{\partial}{\partial x_i}\Big[p_t(\mathbf{x})\nabla\cdot[\mathbf{G}(\mathbf{x},t)\mathbf{G}(\mathbf{x},t)^{\mathsf{T}}] + p_t(\mathbf{x})\mathbf{G}(\mathbf{x},t)\mathbf{G}(\mathbf{x},t)^{\mathsf{T}}\nabla_{\mathbf{x}}\log p_t(\mathbf{x})\Big] \\
&= -\sum_{i=1}^{d} \frac{\partial}{\partial x_i}\Big\{f_i(\mathbf{x},t)p_t(\mathbf{x}) \\
&\quad - \frac{1}{2}\Big[\nabla\cdot[\mathbf{G}(\mathbf{x},t)\mathbf{G}(\mathbf{x},t)^{\mathsf{T}}] + \mathbf{G}(\mathbf{x},t)\mathbf{G}(\mathbf{x},t)^{\mathsf{T}}\nabla_{\mathbf{x}}\log p_t(\mathbf{x})\Big]p_t(\mathbf{x})\Big\} \\
&= -\sum_{i=1}^{d} \frac{\partial}{\partial x_i}[\tilde{f}_i(\mathbf{x},t)p_t(\mathbf{x})], \qquad (37)
\end{aligned}
$$

where we define

$$\tilde{\mathbf{f}}(\mathbf{x},t) := \mathbf{f}(\mathbf{x},t) - \frac{1}{2}\nabla\cdot[\mathbf{G}(\mathbf{x},t)\mathbf{G}(\mathbf{x},t)^{\mathsf{T}}] - \frac{1}{2}\mathbf{G}(\mathbf{x},t)\mathbf{G}(\mathbf{x},t)^{\mathsf{T}}\nabla_{\mathbf{x}}\log p_t(\mathbf{x}).$$

Inspecting Eq. (37), we observe that it equals Kolmogorov's forward equation of the following SDE with $\tilde{\mathbf{G}}(\mathbf{x},t) := \mathbf{0}$ (Kolmogorov's forward equation in this case is also known as the Liouville equation.)

$$\mathrm{d}\mathbf{x} = \tilde{\mathbf{f}}(\mathbf{x},t)\mathrm{d}t + \tilde{\mathbf{G}}(\mathbf{x},t)\mathrm{d}\mathbf{w},$$

which is essentially an ODE:

$$\mathrm{d}\mathbf{x} = \tilde{\mathbf{f}}(\mathbf{x},t)\mathrm{d}t,$$

same as the probability flow ODE given by Eq. (17). Therefore, we have shown that the probability flow ODE Eq. (17) induces the same marginal probability density $p_t(\mathbf{x})$ as the SDE in Eq. (15).

SGMS (cont.)

D.3 Probability flow sampling

Suppose we have a forward SDE

$$\mathrm{d}\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\mathrm{d}t + \mathbf{G}(t)\mathrm{d}\mathbf{w},$$

and one of its discretization

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{f}_i(\mathbf{x}_i) + \mathbf{G}_i\mathbf{z}_i, \quad i = 0, 1, \cdots, N-1, \tag{41}$$

where $\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. We assume the discretization schedule of time is fixed beforehand, and thus we absorb the dependency on Δt into the notations of $\mathbf{f}_i$ and $\mathbf{G}_i$. Using Eq. (17), we can obtain the following probability flow ODE:

$$\mathrm{d}\mathbf{x} = \left\{\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}\mathbf{G}(t)\mathbf{G}(t)^\mathsf{T}\nabla_\mathbf{x} \log p_t(\mathbf{x})\right\} \mathrm{d}t. \tag{42}$$

We may employ any numerical method to integrate the probability flow ODE backwards in time for sample generation. In particular, we propose a discretization in a similar functional form to Eq. (41):

$$\mathbf{x}_i = \mathbf{x}_{i+1} - \mathbf{f}_{i+1}(\mathbf{x}_{i+1}) + \frac{1}{2}\mathbf{G}_{i+1}\mathbf{G}_{i+1}^\mathsf{T}\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}_{i+1}, i+1), \quad i = 0, 1, \cdots, N-1,$$

where the score-based model $\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}_i, i)$ is conditioned on the iteration number i. This is a deterministic iteration rule. Unlike reverse diffusion samplers or ancestral sampling, there is no additional randomness once the initial sample $\mathbf{x}_N$ is obtained from the prior distribution. When applied to SMLD models, we can get the following iteration rule for probability flow sampling:

$$\mathbf{x}_i = \mathbf{x}_{i+1} + \frac{1}{2}(\sigma_{i+1}^2 - \sigma_i^2)\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}_{i+1}, \sigma_{i+1}), \quad i = 0, 1, \cdots, N-1. \tag{43}$$

Similarly, for DDPM models, we have

$$\mathbf{x}_i = (2 - \sqrt{1 - \beta_{i+1}})\mathbf{x}_{i+1} + \frac{1}{2}\beta_{i+1}\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}_{i+1}, i+1), \quad i = 0, 1, \cdots, N-1. \tag{44}$$

SGMS (cont.)

E REVERSE DIFFUSION SAMPLING

Given a forward SDE

$$\mathrm{d}\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\mathrm{d}t + \mathbf{G}(t)\mathrm{d}\mathbf{w},$$

and suppose the following iteration rule is a discretization of it:

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{f}_i(\mathbf{x}_i) + \mathbf{G}_i\mathbf{z}_i, \quad i = 0, 1, \cdots, N-1 \tag{45}$$

where $\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Here we assume the discretization schedule of time is fixed beforehand, and thus we can absorb it into the notations of $\mathbf{f}_i$ and $\mathbf{G}_i$.

Based on Eq. (45), we propose to discretize the reverse-time SDE

$$\mathrm{d}\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - \mathbf{G}(t)\mathbf{G}(t)^{\mathsf{T}}\nabla_{\mathbf{x}} \log p_t(\mathbf{x})]\mathrm{d}t + \mathbf{G}(t)\mathrm{d}\bar{\mathbf{w}},$$

with a similar functional form, which gives the following iteration rule for $i \in \{0, 1, \cdots, N-1\}$:

$$\mathbf{x}_i = \mathbf{x}_{i+1} - \mathbf{f}_{i+1}(\mathbf{x}_{i+1}) + \mathbf{G}_{i+1}\mathbf{G}_{i+1}^{\mathsf{T}}\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}_{i+1}, i+1) + \mathbf{G}_{i+1}\mathbf{z}_{i+1}, \tag{46}$$

where our trained score-based model $\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}_i, i)$ is conditioned on iteration number i.

When applying Eq. (46) to Eqs. (10) and (20), we obtain a new set of numerical solvers for the reverse-time VE and VP SDEs, resulting in sampling algorithms as shown in the "predictor" part of Algorithms 2 and 3. We name these sampling methods (that are based on the discretization strategy in Eq. (46)) *reverse diffusion samplers*.

As expected, the ancestral sampling of DDPM (Ho et al., 2020) (Eq. (4)) matches its reverse diffusion counterpart when $\beta_i \to 0$ for all i (which happens when $\Delta t \to 0$ since $\beta_i = \bar{\beta}_i \Delta t$, see Appendix B),

SGMS (cont.)

because

$$
\begin{aligned}
\mathbf{x}_i &= \frac{1}{\sqrt{1-\beta_{i+1}}}(\mathbf{x}_{i+1} + \beta_{i+1}\mathbf{s}_{\boldsymbol{\theta}*}(\mathbf{x}_{i+1}, i+1)) + \sqrt{\beta_{i+1}}\mathbf{z}_{i+1} \\
&= \left(1 + \frac{1}{2}\beta_{i+1} + o(\beta_{i+1})\right)(\mathbf{x}_{i+1} + \beta_{i+1}\mathbf{s}_{\boldsymbol{\theta}*}(\mathbf{x}_{i+1}, i+1)) + \sqrt{\beta_{i+1}}\mathbf{z}_{i+1} \\
&\approx \left(1 + \frac{1}{2}\beta_{i+1}\right)(\mathbf{x}_{i+1} + \beta_{i+1}\mathbf{s}_{\boldsymbol{\theta}*}(\mathbf{x}_{i+1}, i+1)) + \sqrt{\beta_{i+1}}\mathbf{z}_{i+1} \\
&= \left(1 + \frac{1}{2}\beta_{i+1}\right)\mathbf{x}_{i+1} + \beta_{i+1}\mathbf{s}_{\boldsymbol{\theta}*}(\mathbf{x}_{i+1}, i+1) + \frac{1}{2}\beta_{i+1}^2\mathbf{s}_{\boldsymbol{\theta}*}(\mathbf{x}_{i+1}, i+1) + \sqrt{\beta_{i+1}}\mathbf{z}_{i+1} \\
&\approx \left(1 + \frac{1}{2}\beta_{i+1}\right)\mathbf{x}_{i+1} + \beta_{i+1}\mathbf{s}_{\boldsymbol{\theta}*}(\mathbf{x}_{i+1}, i+1) + \sqrt{\beta_{i+1}}\mathbf{z}_{i+1} \\
&= \left[2 - \left(1 - \frac{1}{2}\beta_{i+1}\right)\right]\mathbf{x}_{i+1} + \beta_{i+1}\mathbf{s}_{\boldsymbol{\theta}*}(\mathbf{x}_{i+1}, i+1) + \sqrt{\beta_{i+1}}\mathbf{z}_{i+1} \\
&\approx \left[2 - \left(1 - \frac{1}{2}\beta_{i+1}\right) + o(\beta_{i+1})\right]\mathbf{x}_{i+1} + \beta_{i+1}\mathbf{s}_{\boldsymbol{\theta}*}(\mathbf{x}_{i+1}, i+1) + \sqrt{\beta_{i+1}}\mathbf{z}_{i+1} \\
&= (2 - \sqrt{1-\beta_{i+1}})\mathbf{x}_{i+1} + \beta_{i+1}\mathbf{s}_{\boldsymbol{\theta}*}(\mathbf{x}_{i+1}, i+1) + \sqrt{\beta_{i+1}}\mathbf{z}_{i+1}.
\end{aligned}
$$

Therefore, the original ancestral sampler of Eq. (4) is essentially a different discretization to the same reverse-time SDE. This unifies the sampling method in Ho et al. (2020) as a numerical solver to the reverse-time VP SDE in our continuous framework.

SGMS (cont.)

F Ancestral sampling for SMLD models

The ancestral sampling method for DDPM models can also be adapted to SMLD models. Consider a sequence of noise scales $\sigma_1 < \sigma_2 < \cdots < \sigma_N$ as in SMLD. By perturbing a data point $\mathbf{x}_0$ with these noise scales sequentially, we obtain a Markov chain $\mathbf{x}_0 \to \mathbf{x}_1 \to \cdots \to \mathbf{x}_N$, where

$$p(\mathbf{x}_i \mid \mathbf{x}_{i-1}) = \mathcal{N}(\mathbf{x}_i; \mathbf{x}_{i-1}, (\sigma_i^2 - \sigma_{i-1}^2)\mathbf{I}), \quad i = 1, 2, \cdots, N.$$

Here we assume $\sigma_0 = 0$ to simplify notations. Following Ho et al. (2020), we can compute

$$q(\mathbf{x}_{i-1} \mid \mathbf{x}_i, \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_{i-1}; \frac{\sigma_{i-1}^2}{\sigma_i^2}\mathbf{x}_i + \left(1 - \frac{\sigma_{i-1}^2}{\sigma_i^2}\right)\mathbf{x}_0, \frac{\sigma_{i-1}^2(\sigma_i^2 - \sigma_{i-1}^2)}{\sigma_i^2}\mathbf{I}\right).$$

If we parameterize the reverse transition kernel as $p_{\boldsymbol{\theta}}(\mathbf{x}_{i-1} \mid \mathbf{x}_i) = \mathcal{N}(\mathbf{x}_{i-1}; \boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{x}_i, i), \tau_i^2\mathbf{I})$, then

$$\begin{aligned} L_{t-1} &= \mathbb{E}_q[D_{\mathrm{KL}}(q(\mathbf{x}_{i-1} \mid \mathbf{x}_i, \mathbf{x}_0)) \parallel p_{\boldsymbol{\theta}}(\mathbf{x}_{i-1} \mid \mathbf{x}_i)] \\ &= \mathbb{E}_q\left[\frac{1}{2\tau_i^2}\left\| \frac{\sigma_{i-1}^2}{\sigma_i^2}\mathbf{x}_i + \left(1 - \frac{\sigma_{i-1}^2}{\sigma_i^2}\right)\mathbf{x}_0 - \boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{x}_i, i)\right\|_2^2\right] + C \\ &= \mathbb{E}_{\mathbf{x}_0, \mathbf{z}}\left[\frac{1}{2\tau_i^2}\left\|\mathbf{x}_i(\mathbf{x}_0, \mathbf{z}) - \frac{\sigma_i^2 - \sigma_{i-1}^2}{\sigma_i}\mathbf{z} - \boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{x}_i(\mathbf{x}_0, \mathbf{z}), i)\right\|_2^2\right] + C, \end{aligned}$$

where L_{t-1} is one representative term in the ELBO objective (see Eq. (8) in Ho et al. (2020)), C is a constant that does not depend on $\boldsymbol{\theta}$, $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and $\mathbf{x}_i(\mathbf{x}_0, \mathbf{z}) = \mathbf{x}_0 + \sigma_i\mathbf{z}$. We can therefore parameterize $\boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{x}_i, i)$ via

$$\boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{x}_i, i) = \mathbf{x}_i + (\sigma_i^2 - \sigma_{i-1}^2)\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_i, i),$$

where $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_i, i)$ is to estimate $\mathbf{z}/\sigma_i$. As in Ho et al. (2020), we let $\tau_i = \sqrt{\frac{\sigma_{i-1}^2(\sigma_i^2 - \sigma_{i-1}^2)}{\sigma_i^2}}$. Through ancestral sampling on $\prod_{i=1}^N p_{\boldsymbol{\theta}}(\mathbf{x}_{i-1} \mid \mathbf{x}_i)$, we obtain the following iteration rule

$$\mathbf{x}_{i-1} = \mathbf{x}_i + (\sigma_i^2 - \sigma_{i-1}^2)\mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}_i, i) + \sqrt{\frac{\sigma_{i-1}^2(\sigma_i^2 - \sigma_{i-1}^2)}{\sigma_i^2}}\mathbf{z}_i, i = 1, 2, \cdots, N, \tag{47}$$

where $\mathbf{x}_N \sim \mathcal{N}(\mathbf{0}, \sigma_N^2\mathbf{I})$, $\boldsymbol{\theta}^*$ denotes the optimal parameter of $\mathbf{s}_{\boldsymbol{\theta}}$, and $\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. We call Eq. (47) the ancestral sampling method for SMLD models.

SGMS (cont.)

I CONTROLLABLE GENERATION

Consider a forward SDE with the following general form

$$\mathrm{d}\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\mathrm{d}t + \mathbf{G}(\mathbf{x}, t)\mathrm{d}\mathbf{w},$$

and suppose the initial state distribution is $p_0(\mathbf{x}(0) \mid \mathbf{y})$. The density at time t is $p_t(\mathbf{x}(t) \mid \mathbf{y})$ when conditioned on $\mathbf{y}$. Therefore, using Anderson (1982), the reverse-time SDE is given by

$$\mathrm{d}\mathbf{x} = \{\mathbf{f}(\mathbf{x}, t) - \nabla \cdot [\mathbf{G}(\mathbf{x}, t)\mathbf{G}(\mathbf{x}, t)^{\mathsf{T}}] - \mathbf{G}(\mathbf{x}, t)\mathbf{G}(\mathbf{x}, t)^{\mathsf{T}}\nabla_{\mathbf{x}} \log p_t(\mathbf{x} \mid \mathbf{y})\}\mathrm{d}t + \mathbf{G}(\mathbf{x}, t)\mathrm{d}\bar{\mathbf{w}}. \quad (48)$$

Since $p_t(\mathbf{x}(t) \mid \mathbf{y}) \propto p_t(\mathbf{x}(t))p(\mathbf{y} \mid \mathbf{x}(t))$, the score $\nabla_{\mathbf{x}} \log p_t(\mathbf{x}(t) \mid \mathbf{y})$ can be computed easily by

$$\nabla_{\mathbf{x}} \log p_t(\mathbf{x}(t) \mid \mathbf{y}) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x}(t)) + \nabla_{\mathbf{x}} \log p(\mathbf{y} \mid \mathbf{x}(t)). \quad (49)$$

This subsumes the conditional reverse-time SDE in Eq. (14) as a special case. All sampling methods we have discussed so far can be applied to the conditional reverse-time SDE for sample generation.

For imputation, our goal is to sample from $p(\bar{\Omega}(\mathbf{x}(0)) \mid \Omega(\mathbf{x}(0)) = \mathbf{y})$. Define a new diffusion process $\mathbf{z}(t) = \bar{\Omega}(\mathbf{x}(t))$, and note that the SDE for $\mathbf{z}(t)$ can be written as

$$\mathrm{d}\mathbf{z} = \mathbf{f}_{\bar{\Omega}}(\mathbf{z}, t)\mathrm{d}t + \mathbf{G}_{\bar{\Omega}}(\mathbf{z}, t)\mathrm{d}\mathbf{w}.$$

The reverse-time SDE, conditioned on $\Omega(\mathbf{x}(0)) = \mathbf{y}$, is given by

$$\begin{aligned}\mathrm{d}\mathbf{z} = \{&\mathbf{f}_{\bar{\Omega}}(\mathbf{z}, t) - \nabla \cdot [\mathbf{G}_{\bar{\Omega}}(\mathbf{z}, t)\mathbf{G}_{\bar{\Omega}}(\mathbf{z}, t)^{\mathsf{T}}]\\ &- \mathbf{G}_{\bar{\Omega}}(\mathbf{z}, t)\mathbf{G}_{\bar{\Omega}}(\mathbf{z}, t)^{\mathsf{T}}\nabla_{\mathbf{z}} \log p_t(\mathbf{z} \mid \Omega(\mathbf{z}(0)) = \mathbf{y})\}\mathrm{d}t + \mathbf{G}_{\bar{\Omega}}(\mathbf{z}, t)\mathrm{d}\bar{\mathbf{w}}.\end{aligned}$$

Although $p_t(\mathbf{z}(t) \mid \Omega(\mathbf{x}(0)) = \mathbf{y})$ is in general intractable, it can be approximated. Let A denote the event $\Omega(\mathbf{x}(0)) = \mathbf{y}$. We have

$$\begin{aligned}p_t(\mathbf{z}(t) \mid \Omega(\mathbf{x}(0)) = \mathbf{y}) = p_t(\mathbf{z}(t) \mid A) &= \int p_t(\mathbf{z}(t) \mid \Omega(\mathbf{x}(t)), A)p_t(\Omega(\mathbf{x}(t)) \mid A)\mathrm{d}\Omega(\mathbf{x}(t))\\ &= \mathbb{E}_{p_t(\Omega(\mathbf{x}(t))|A)}[p_t(\mathbf{z}(t) \mid \Omega(\mathbf{x}(t)), A)]\\ &\approx \mathbb{E}_{p_t(\Omega(\mathbf{x}(t))|A)}[p_t(\mathbf{z}(t) \mid \Omega(\mathbf{x}(t)))]\\ &\approx p_t(\mathbf{z}(t) \mid \hat{\Omega}(\mathbf{x}(t))),\end{aligned}$$

where $\hat{\Omega}(\mathbf{x}(t))$ is a random sample from $p_t(\Omega(\mathbf{x}(t)) \mid A)$, which is typically a tractable distribution. Therefore,

$$\begin{aligned}\nabla_{\mathbf{z}} \log p_t(\mathbf{z}(t) \mid \Omega(\mathbf{x}(0)) = \mathbf{y}) &\approx \nabla_{\mathbf{z}} \log p_t(\mathbf{z}(t) \mid \hat{\Omega}(\mathbf{x}(t)))\\ &= \nabla_{\mathbf{z}} \log p_t([\mathbf{z}(t); \hat{\Omega}(\mathbf{x}(t))]),\end{aligned}$$

where $[\mathbf{z}(t); \hat{\Omega}(\mathbf{x}(t))]$ denotes a vector $\mathbf{u}(t)$ such that $\Omega(\mathbf{u}(t)) = \hat{\Omega}(\mathbf{x}(t))$ and $\bar{\Omega}(\mathbf{u}(t)) = \mathbf{z}(t)$, and the identity holds because $\nabla_{\mathbf{z}} \log p_t([\mathbf{z}(t); \hat{\Omega}(\mathbf{x}(t))]) = \nabla_{\mathbf{z}} \log p_t(\mathbf{z}(t) \mid \hat{\Omega}(\mathbf{x}(t))) + \nabla_{\mathbf{z}} \log p(\hat{\Omega}(\mathbf{x}(t))) = \nabla_{\mathbf{z}} \log p_t(\mathbf{z}(t) \mid \hat{\Omega}(\mathbf{x}(t)))$.

We provided an extended set of inpainting results in Figs. 14 and 15.

SGMS (cont.)

I.4 Solving general inverse problems

Suppose we have two random variables $\mathbf{x}$ and $\mathbf{y}$, and we know the forward process of generating $\mathbf{y}$ from $\mathbf{x}$, given by $p(\mathbf{y} \mid \mathbf{x})$. The inverse problem is to obtain $\mathbf{x}$ from $\mathbf{y}$, that is, generating samples from $p(\mathbf{x} \mid \mathbf{y})$. In principle, we can estimate the prior distribution $p(\mathbf{x})$ and obtain $p(\mathbf{x} \mid \mathbf{y})$ using Bayes' rule: $p(\mathbf{x} \mid \mathbf{y}) = p(\mathbf{x})p(\mathbf{y} \mid \mathbf{x})/p(\mathbf{y})$. In practice, however, both estimating the prior and performing Bayesian inference are non-trivial.

Leveraging Eq. (48), score-based generative models provide one way to solve the inverse problem. Suppose we have a diffusion process $\{\mathbf{x}(t)\}_{t=0}^{T}$ generated by perturbing $\mathbf{x}$ with an SDE, and a

time-dependent score-based model $\mathbf{s}_{\boldsymbol{\theta}*}(\mathbf{x}(t), t)$ trained to approximate $\nabla_{\mathbf{x}} \log p_t(\mathbf{x}(t))$. Once we have an estimate of $\nabla_{\mathbf{x}} \log p_t(\mathbf{x}(t) \mid \mathbf{y})$, we can simulate the reverse-time SDE in Eq. (48) to sample from $p_0(\mathbf{x}(0) \mid \mathbf{y}) = p(\mathbf{x} \mid \mathbf{y})$. To obtain this estimate, we first observe that

$$\nabla_{\mathbf{x}} \log p_t(\mathbf{x}(t) \mid \mathbf{y}) = \nabla_{\mathbf{x}} \log \int p_t(\mathbf{x}(t) \mid \mathbf{y}(t), \mathbf{y}) p(\mathbf{y}(t) \mid \mathbf{y}) \mathrm{d}\mathbf{y}(t),$$

where $\mathbf{y}(t)$ is defined via $\mathbf{x}(t)$ and the forward process $p(\mathbf{y}(t) \mid \mathbf{x}(t))$. Now assume two conditions:

- $p(\mathbf{y}(t) \mid \mathbf{y})$ is tractable. We can often derive this distribution from the interaction between the forward process and the SDE, like in the case of image imputation and colorization.
- $p_t(\mathbf{x}(t) \mid \mathbf{y}(t), \mathbf{y}) \approx p_t(\mathbf{x}(t) \mid \mathbf{y}(t))$. For small t, $\mathbf{y}(t)$ is almost the same as $\mathbf{y}$ so the approximation holds. For large t, $\mathbf{y}$ becomes further away from $\mathbf{x}(t)$ in the Markov chain, and thus have smaller impact on $\mathbf{x}(t)$. Moreover, the approximation error for large t matter less for the final sample, since it is used early in the sampling process.

Given these two assumptions, we have

$$\begin{aligned}
\nabla_{\mathbf{x}} \log p_t(\mathbf{x}(t) \mid \mathbf{y}) &\approx \nabla_{\mathbf{x}} \log \int p_t(\mathbf{x}(t) \mid \mathbf{y}(t)) p(\mathbf{y}(t) \mid \mathbf{y}) \mathrm{d}\mathbf{y} \\
&\approx \nabla_{\mathbf{x}} \log p_t(\mathbf{x}(t) \mid \hat{\mathbf{y}}(t)) \\
&= \nabla_{\mathbf{x}} \log p_t(\mathbf{x}(t)) + \nabla_{\mathbf{x}} \log p_t(\hat{\mathbf{y}}(t) \mid \mathbf{x}(t)) \\
&\approx \mathbf{s}_{\boldsymbol{\theta}*}(\mathbf{x}(t), t) + \nabla_{\mathbf{x}} \log p_t(\hat{\mathbf{y}}(t) \mid \mathbf{x}(t)), \qquad (50)
\end{aligned}$$

where $\hat{\mathbf{y}}(t)$ is a sample from $p(\mathbf{y}(t) \mid \mathbf{y})$. Now we can plug Eq. (50) into Eq. (48) and solve the resulting reverse-time SDE to generate samples from $p(\mathbf{x} \mid \mathbf{y})$.

Diffusion based generative modeling - Optimal control perspective

OCDBGM (cont.)

2 SDE-based generative modeling as an optimal control problem

Diffusion models can naturally be interpreted through the lens of continuous-time stochastic processes (Song et al., 2021). To this end, let us formalize our setting in the general context of SDE-based generative models. We define our model as the stochastic process $X = (X_s)_{s\in[0,T]}$ characterized by the SDE

$$\mathrm{d}X_s = \bar{\mu}(X_s, s)\,\mathrm{d}s + \bar{\sigma}(s)\,\mathrm{d}B_s \tag{1}$$

with suitable[2] drift and diffusion coefficients $\mu\colon \mathbb{R}^d \times [0,T] \to \mathbb{R}^d$ and $\sigma\colon [0,T] \to \mathbb{R}^{d\times d}$. Learning the model in (1) now corresponds to solving the following problem.

Problem 2.1 (SDE-based generative modeling)**.** Learn an initial condition X_0 as well as coefficient functions μ and σ such that the distribution of X_T approximates a given data distribution $\mathcal{D}$.

While, in general, the initial condition X_0 as well as both the coefficient functions μ and σ can be learned, typical applications often focus on learning only the drift μ (Ho et al., 2020; Kong et al., 2021; Corso et al., 2022). The following remark justifies that this is sufficient to represent an arbitrary distribution $\mathcal{D}$.

Remark 2.2 (Reverse-time SDE). Naively, one can achieve $X_T \sim \mathcal{D}$ by setting

$$X_0 \sim Y_T \quad \text{and} \quad \mu := \sigma\sigma^\top \nabla \log p_Y - f, \tag{2}$$

where $f\colon \mathbb{R}^d \times [0,T] \to \mathbb{R}^d$ and Y is a solution to the SDE

$$\mathrm{d}Y_s = f(Y_s, s)\,\mathrm{d}s + \sigma(s)\,\mathrm{d}B_s, \quad Y_0 \sim \mathcal{D}. \tag{3}$$

This well-known result dates back to Nelson (1967); Anderson (1982); Haussmann & Pardoux (1986); Föllmer (1988). More generally, it states that X can be interpreted as the time-reversal of Y, which we will denote by $\bar{Y}$, in the sense that $p_X = \bar{p}_Y$ almost everywhere, see Figure 2 and Section A.3. Even though the reverse-time SDE provides a valid solution to Problem 2.1, apparent practical challenges are to sample from the terminal distribution of the generative SDE, $X_0 \sim Y_T$, and to compute the so-called *score* $\nabla \log p_Y$. In most scenarios, we either only have access to samples from the true data distribution $\mathcal{D}$ or access to its density. If samples from $\mathcal{D}$ are available, diffusion models provide a working solution to solving Problem 2.1 (more details in Section 2.4). When, instead, having access to the density of $\mathcal{D}$, general time-reversed diffusions have, to the best of our knowledge, not yet been considered. In Section 3, we thus propose a novel strategy for this scenario.

OCDBGM (cont.)

2.1 PDE perspective: HJB equation for log-density

We start with the well-known *Fokker-Planck equation*, which describes the evolution of the density of the solution X to the SDE in (1) via the PDE

$$\partial_t p_X = \operatorname{div}\left(\operatorname{div}\left(\bar{D} p_X\right) - \bar{\mu} p_X\right), \tag{4}$$

where we set $D := \frac{1}{2}\sigma\sigma^\top$ for notational convenience. This implies that the time-reversed density $\bar{p}_X$ satisfies a (generalized) *Kolmogorov backward equation* given by

$$\partial_t \bar{p}_X = \operatorname{div}\left(-\operatorname{div}\left(D\bar{p}_X\right) + \mu\bar{p}_X\right) = -\operatorname{Tr}\left(D\nabla^2\bar{p}_X\right) + \mu\cdot\nabla\bar{p}_X + \operatorname{div}(\mu)\bar{p}_X. \tag{5}$$

The second equality follows from the identities for divergences in Section A.2 and the fact that σ does not depend on the spatial variable[3] x. Now we use the *Hopf–Cole transformation* to convert the linear PDE in (5) to an HJB equation which is prominent in control theory, see Pavon (1989); Fleming & Rishel (2012), and Section A.5.

Lemma 2.3 (HJB equation for log-density)**.** *Let us define $V := -\log\bar{p}_X$. Then V is a solution to the HJB equation*

$$\partial_t V = -\operatorname{Tr}\left(D\nabla^2 V\right) + \mu\cdot\nabla V - \operatorname{div}(\mu) + \frac{1}{2}\left\|\sigma^\top\nabla V\right\|^2 \tag{6}$$

with terminal condition $V(\cdot, T) = -\log p_{X_0}$.

OCDBGM (cont.)

2.2 Optimal control perspective: ELBO derivation

We will derive the ELBO for our generative model (1) using the following fundamental result from control theory, which shows that the solution to an HJB equation, such as the one stated in Lemma 2.3, is related to an optimal control problem, see Dai Pra (1991); Pavon (1989); Nüsken & Richter (2021); Fleming & Soner (2006); Pham (2009) and Section A.7. For a general introduction to stochastic optimal control theory we refer to Section A.6.

Theorem 2.4 (Verification theorem). *Let V be a solution to the HJB equation in Lemma 2.3. Further, let $\mathcal{U} \subset C^1(\mathbb{R}^d \times [0,T], \mathbb{R}^d)$ be a suitable set of admissible controls and for every control $u \in \mathcal{U}$ let Y^u be the solution to the controlled SDE*[4]

$$\mathrm{d}Y_s^u = (\sigma u - \mu)(Y_s^u, s)\,\mathrm{d}s + \sigma(s)\,\mathrm{d}B_s. \tag{7}$$

Then it holds almost surely that

$$V(Y_0^u, 0) = \min_{u \in \mathcal{U}} \mathbb{E}\left[\mathcal{R}_\mu^u(Y^u) - \log p_{X_0}(Y_T^u) \middle| Y_0^u\right], \tag{8}$$

where $-\log p_{X_0}(Y_T^u)$ constitutes the terminal costs, and the running costs are defined as

$$\mathcal{R}_\mu^u(Y^u) := \int_0^T \left(\operatorname{div}(\mu) + \frac{1}{2}\|u\|^2\right)(Y_s^u, s)\,\mathrm{d}s. \tag{9}$$

Moreover, the unique minimum is attained by $u^* := -\sigma^\top \nabla V$.

Proposition 2.6 (Optimal path space measure). *The optimal path space measure can be defined via the work functional $\mathcal{W}\colon C([0,T], \mathbb{R}^d) \to \mathbb{R}$ and the Radon-Nikodym derivative*

$$\frac{\mathrm{d}\mathbb{P}_{Y^{u^*}}}{\mathrm{d}\mathbb{P}_{Y^0}} = \exp(-\mathcal{W}) \quad \text{with} \quad \mathcal{W}(Y^0) := \mathcal{R}_\mu^0(Y^0) - \log \frac{p_{X_0}(Y_T^0)}{p_{X_T}(Y_0^0)}, \tag{11}$$

where $\mathcal{R}_\mu^0(Y^0)$ is as in (9) with $u = 0$. Moreover, for any $u \in \mathcal{U}$, the expected variational gap

$$G(u) := \mathbb{E}\left[\log p_{X_T}(Y_0^u)\right] - \mathbb{E}\left[\log p_{X_0}(Y_T^u) - \mathcal{R}_\mu^u(Y^u)\right] \tag{12}$$

of the ELBO in Corollary 2.5 satisfies that

$$G(u) = D_{\mathrm{KL}}(\mathbb{P}_{Y^u}|\mathbb{P}_{Y^{u^*}}) = D_{\mathrm{KL}}(\mathbb{P}_{Y^u}|\mathbb{P}_{\bar{X}}) - D_{\mathrm{KL}}(\mathbb{P}_{Y_0^u}|\mathbb{P}_{X_T}). \tag{13}$$

OCDBGM (cont.)

2.4 Connection to denoising score matching objective

This section outlines that, under a reparametrization of the generative model in (1), the ELBO in Corollary 2.5 corresponds to the objective typically used for the training of continuous-time diffusion models. We note that the ELBO in Corollary 2.5 in fact equals the one derived in Huang et al. (2021, Theorem 3). Following the arguments therein and motivated by Remark 2.2, we can now use the reparametrization $\mu := \sigma u - f$ to arrive at an uncontrolled *inference SDE* and a controlled *generative SDE*

$$\mathrm{d}Y_s = f(Y_s, s)\,\mathrm{d}s + \sigma(s)\,\mathrm{d}B_s, \quad Y_0 \sim \mathcal{D}, \quad \text{and} \quad \mathrm{d}X_s^u = \left(\bar{\sigma}\bar{u} - \bar{f}\right)(X_s^u, s)\,\mathrm{d}s + \bar{\sigma}(s)\,\mathrm{d}B_s. \tag{14}$$

In practice, the coefficients f and σ are usually[6] constructed such that Y is an Ornstein-Uhlenbeck (OU) process with Y_T being approximately distributed according to a standard normal distribution, see also (139) in Section A.13. This is why the process Y is said to *diffuse* the data. Setting $X_0^u \sim \mathcal{N}(0, \mathrm{I})$ thus satisfies that $p_{X_0^u} \approx p_{Y_T}$ and allows to easily sample X_0^u.

The corresponding ELBO in Corollary 2.5 now takes the form

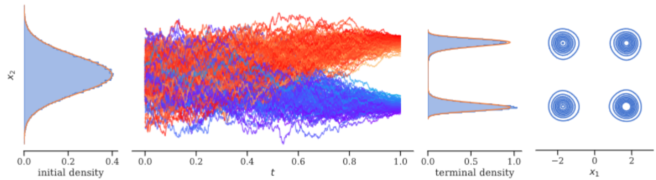
$$\log p_{X_T^u}(Y_0) \geq \mathbb{E}\left[\log p_{X_0^u}(Y_T) - \mathcal{R}^u_{\sigma u - f}(Y) \middle| Y_0\right], \tag{15}$$

where, in analogy to (13), the expected variational gap is given by the forward KL divergence $G(u) = D_{\mathrm{KL}}(\mathbb{P}_Y|\mathbb{P}_{\bar{X}^u}) - D_{\mathrm{KL}}(\mathbb{P}_{Y_0}|\mathbb{P}_{X_T^u})$, see Proposition A.10. We note that the process Y in the ELBO does not depend on the control u anymore. Under suitable assumptions, this allows us to rewrite the expected negative ELBO (up to a constant not depending on u) as a denoising score matching objective (Vincent, 2011), i.e.,

$$\mathcal{L}_{\mathrm{DSM}}(u) := \frac{T}{2}\mathbb{E}\left[\left\|u(Y_\tau, \tau) - \sigma^\top(\tau)\nabla \log p_{Y_\tau|Y_0}(Y_\tau|Y_0)\right\|^2\right], \tag{16}$$

where $\tau \sim \mathcal{U}([0, T])$ and $p_{Y_\tau|Y_0}$ denotes the conditional density of Y_τ given Y_0, see Section A.9. We emphasize that the conditional density can be explicitly computed for the OU process, see also (137) in Section A.13. Due to its simplicity, variants of the objective (16) are typically used in implementations. Note, however, that the setting in this section requires that one has access to samples of $\mathcal{D}$ in order to simulate the process Y. In the next section, we consider a different scenario, where instead we only have access to the (unnormalized) density of $\mathcal{D}$.

OCDBGM (cont.)



Corollary 3.1 (Reverse KL divergence). *Let X^u and Y be defined by (17). Then it holds*

$$D_{\mathrm{KL}}(\mathbb{P}_{X^u}|\mathbb{P}_{\tilde{Y}}) = D_{\mathrm{KL}}(\mathbb{P}_{X^u}|\mathbb{P}_{X^{u^*}}) + D_{\mathrm{KL}}(\mathbb{P}_{X_0^u}|\mathbb{P}_{Y_T}) \tag{18a}$$

$$= \mathbb{E}\left[\mathcal{R}_{\tilde{f}}^{\tilde{u}}(X^u) + \log\frac{p_{Y_T}(X_0^u)}{\rho(X_T^u)}\right] + \log\mathcal{Z} + D_{\mathrm{KL}}(\mathbb{P}_{X_0^u}|\mathbb{P}_{Y_T}) \tag{18b}$$

$$= \mathcal{L}_{\mathrm{DIS}}(u) + \log\mathcal{Z}, \tag{18c}$$

where

$$\mathcal{L}_{\mathrm{DIS}}(u) := \mathbb{E}\left[\mathcal{R}_{\tilde{f}}^{\tilde{u}}(X^u) + \log\frac{p_{X_0^u}(X_0^u)}{\rho(X_T^u)}\right]. \tag{19}$$

In particular, this implies that

$$D_{\mathrm{KL}}(\mathbb{P}_{X_0^u}|\mathbb{P}_{Y_T}) - \log\mathcal{Z} = \min_{u\in\mathcal{U}} \mathcal{L}_{\mathrm{DIS}}(u) \tag{20}$$

and, assuming that $X_0^u \sim Y_T$, the minimizing control $u^* := \sigma^\top \nabla \log p_Y$ guarantees that $X_T^{u^*} \sim \mathcal{D}$.

OCDBGM (cont.)

Theorem A.1 (Reverse-time SDE). *Let $f \in \mathbb{R}^d \times [0,T] \to \mathbb{R}^d$ and $\sigma\colon \mathbb{R}^d \times [0,T] \to \mathbb{R}^{d\times d}$, let $Y = (Y_s)_{s\in[0,T]}$ be the solution to the SDE*

$$\mathrm{d}Y_s = f(Y_s, s)\,\mathrm{d}s + \sigma(Y_s, s)\,\mathrm{d}B_s, \tag{33}$$

and assume that Y has density p_Y, which satisfies the Fokker-Planck equation given by

$$\partial_t p_Y = \operatorname{div}\left(\operatorname{div}\left(Dp_Y\right) - fp_Y\right), \tag{34}$$

where $D := \frac{1}{2}\sigma\sigma^\top$. For every $\lambda \in C^2([0,T],[0,1])$ the solution $\overleftarrow{Y} = (\overleftarrow{Y}_s)_{s\in[0,T]}$ to the reverse-time SDE

$$\mathrm{d}\overleftarrow{Y}_s = \overleftarrow{\mu}^{(\lambda)}(\overleftarrow{Y}_s, s)\,\mathrm{d}s + \overleftarrow{\sigma}^{(\lambda)}(\overleftarrow{Y}_s, s)\,\mathrm{d}B_s, \quad \overleftarrow{Y}_0 \sim Y_T, \tag{35}$$

with

$$\mu^{(\lambda)} := (2-\lambda)\operatorname{div}\left(D\right) + (2-\lambda)D\nabla\log p_Y - f, \tag{36}$$

and

$$\sigma^{(\lambda)} := \sqrt{1-\lambda}\sigma \tag{37}$$

has density $p_{\overleftarrow{Y}}$ given by

$$p_{\overleftarrow{Y}}(\cdot, t) = \overleftarrow{p}_Y(\cdot, t) \tag{38}$$

almost everywhere for every $t \in [0,T]$. In other words, for every $t \in [0,T]$ it holds that $Y_{T-t} \sim \overleftarrow{Y}_t$.

OCDBGM (cont.)

In order to solve Problem 2.1, one might be tempted to rely on classical methods to approximate the solution of the HJB equation from Lemma 2.3 directly. However, in the setting of Remark 2.2, one should note that the optimal drift,

$$\mu = \sigma\sigma^\top \nabla \log p_Y - f = -\sigma\sigma^\top \nabla V - f, \tag{42}$$

contains the solution V itself. Plugging it into the HJB equation in Lemma 2.3, we get the equation

$$\partial_t V = \operatorname{Tr}\left(D\nabla^2 V\right) - f \cdot \nabla V + \operatorname{div}(f) - \frac{1}{2}\|\sigma^\top \nabla V\|^2, \quad V(\cdot, T) = -\log p_{X_0}. \tag{43}$$

Likewise, when applying the Hopf–Cole transformation from Section A.5 to p_Y directly, i.e., considering $V := -\log p_Y$, where Y is the solution to SDE (3), we get the same PDE. We note that the signs in (43) do not match with typical HJB equations from control theory. In order to obtain an HJB equation, we can consider the time-reversed function $\bar{V}$, which satisfies

$$\partial_t \bar{V} = -\operatorname{Tr}\left(\bar{D}\nabla^2 \bar{V}\right) + \bar{f} \cdot \nabla \bar{V} - \operatorname{div}(\bar{f}) + \frac{1}{2}\|\bar{\sigma}^\top \nabla \bar{V}\|^2, \quad \bar{V}(\cdot, T) = -\log p_{X_T}. \tag{44}$$

Recall that the Kolmogorov backward equation follows from the Fokker-Planck equation by using the divergence identities from Section A.2, i.e.

$$\partial_t \bar{p}_X = \operatorname{div}\left(-\operatorname{div}\left(D\bar{p}_X\right) + \mu\bar{p}_X\right) = \operatorname{div}\left(-D\nabla\bar{p}_X\right) + \mu \cdot \nabla\bar{p}_X + \operatorname{div}(\mu)\bar{p}_X \tag{45a}$$

$$= -\operatorname{Tr}\left(D\nabla^2\bar{p}_X\right) + \mu \cdot \nabla\bar{p}_X + \operatorname{div}(\mu)\bar{p}_X. \tag{45b}$$

The following lemma details the relation of the HJB equation in Lemma 2.3 and the linear Kolmogorov backward equation[9] in (5). A proof of the Hopf-Cole transformation can, e.g., be found in Evans (2010, Section 4.4.1) and Richter & Berner (2022, Appendix G). Lemma 2.3 follows with the choices $b := -\mu$ and $h := \operatorname{div}(\mu)$.

Lemma A.2 (Hopf–Cole transformation). *Let $h\colon \mathbb{R}^d \times [0,T] \to \mathbb{R}$ and let $p \in C^{2,1}(\mathbb{R}^d \times [0,T], \mathbb{R})$ solve the linear PDE*

$$\partial_t p = -\frac{1}{2}\operatorname{Tr}(\sigma\sigma^\top \nabla^2 p) - b \cdot \nabla p + hp. \tag{46}$$

Then $V := -\log p$ satisfies the HJB equation

$$\partial_t V = -\frac{1}{2}\operatorname{Tr}(\sigma\sigma^\top \nabla^2 V) - b \cdot \nabla V - h + \frac{1}{2}\left\|\sigma^\top \nabla V\right\|^2. \tag{47}$$

OCDBGM (cont.)

A.6 Brief introduction to stochastic optimal control

In this section, we shall provide a brief introduction to stochastic optimal control. For details and further reading, we refer the interested reader to the monographs by Fleming & Rishel (2012); Fleming & Soner (2006); Pham (2009); Van Handel (2007). Loosely speaking, stochastic control theory deals with identifying optimal strategies in noisy environments, in our case, continuous-time stochastic processes defined by the SDE

$$\mathrm{d}X_s^U = \widetilde{\mu}(X_s^U, s, U_s)\,\mathrm{d}s + \widetilde{\sigma}(X_s^U, s, U_s)\,\mathrm{d}B_s, \tag{48}$$

where $\widetilde{\mu}$ and $\widetilde{\sigma}$ are suitable functions, B is a d-dimensional Brownian motion, and U is a progressively measurable, $\mathbb{R}^d$-valued random control process. For ease of presentation, we focus on the frequent case where U is a *Markov control*, which means that there exists a deterministic function $u \in \mathcal{U} \subset C(\mathbb{R}^d \times [0,T], \mathbb{R}^d)$, such that $U_s = u(X_s^U, s)$. In other words, the randomness of the process U is only coming from the stochastic process X^U. The function class $\mathcal{U}$ then defines the set of *admissible controls*. Very often, one considers the special cases

$$\widetilde{\mu}(x, s, u) := \mu(x, s) + \sigma(s)u(x, s) \qquad \text{and} \qquad \widetilde{\sigma}(x, s, u) := \sigma(s), \tag{49}$$

where μ and σ might correspond to the choices taken in Section 2.4, and one might think of u as a steering force as to reach a certain target.

The goal is now to minimize specified control costs with respect to the control u. To this end, we can define the cost functional

$$J(u; x_{\text{init}}, 0) = \mathbb{E}\left[\int_0^T \widetilde{h}(X_s^U, s, u(X_s^U, s))\,\mathrm{d}s + g(X_T^U) \middle| X_0^U = x_{\text{init}}\right], \tag{50}$$

where $\widetilde{h} : \mathbb{R}^d \times [0,T] \times \mathbb{R}^d \to \mathbb{R}$ specifies *running costs* and $g : \mathbb{R}^d \to \mathbb{R}$ represents *terminal costs*. Furthermore we can define the *cost-to-go* as

$$J(u; x, t) = \mathbb{E}\left[\int_t^T \widetilde{h}(X_s^U, s, u(X_s^U, s))\,\mathrm{d}s + g(X_T^U) \middle| X_t^U = x\right], \tag{51}$$

now depending on respective initial values $(x,t) \in \mathbb{R}^d \times [0,T]$. The objective in optimal control is now to minimize this quantity over all admissible controls $u \in \mathcal{U}$ and we, therefore, introduce the so-called *value function*

$$V(x, t) = \inf_{u \in \mathcal{U}} J(u; x, t) \tag{52}$$

as the optimal costs conditioned on being in position x at time t.

OCDBGM (cont.)

Theorem A.3 (Verification theorem for general HJB equation). *Let $V \in C^{2,1}(\mathbb{R}^d \times [0,T], \mathbb{R})$ fulfill the PDE*

$$\partial_t V = -\inf_{\alpha \in \mathbb{R}^d} \left\{ \widetilde{h}(\cdot,\cdot,\alpha) + \widetilde{\mu}(\cdot,\cdot,\alpha) \cdot \nabla V + \frac{1}{2} \operatorname{Tr}\left((\widetilde{\sigma}\widetilde{\sigma}^\top)(\cdot,\cdot,\alpha) \nabla^2 V \right) \right\}, \quad V(\cdot, T) = g, \tag{53}$$

such that

$$\sup_{(x,t) \in \mathbb{R}^d \times [0,T]} \frac{\|V(x,t)\|}{1 + \|x\|^2} < \infty \tag{54}$$

and assume that there exists a measurable function $\mathcal{U} \ni u^* : \mathbb{R}^d \times [0,T] \to \mathbb{R}^d$ that attains the above infimum for all $(x,t) \in \mathbb{R}^d \times [0,T]$. Further, let the correspondingly controlled SDE in (48) with $U_s^* := u^*(X_s^{U^*}, s)$ have a strong solution X^{U^*}. Then V coincides with the value function as defined in (52) and u^* is an optimal Markovian control.

Corollary A.4 (HJB equation with quadratic running costs). *If the diffusion coefficient $\widetilde{\sigma}$ does not depend on the control, the control enters additively in the drift as in (49), and the running costs take the form*

$$\widetilde{h}(x, s, u(x,s)) = h(x,s) + \frac{1}{2}\|u(x,s)\|^2, \tag{56}$$

then the general HJB equation in (53) can be stated in closed form as the HJB equation in (47).

Proof. We formally compute

$$\inf_{\alpha \in \mathbb{R}^d} \left\{ \widetilde{h}(\cdot,\cdot,\alpha) + \widetilde{\mu}(\cdot,\cdot,\alpha) \cdot \nabla V \right\} = h + \mu \cdot \nabla V + \inf_{\alpha \in \mathbb{R}^d} \left\{ \frac{1}{2}\|\alpha\|^2 + \sigma\alpha \cdot \nabla V \right\}, \tag{57}$$

and realize that the infimum is attained when choosing $\alpha^* = -\sigma^\top \nabla V(x,t)$ for each corresponding $(x,t) \in \mathbb{R}^d \times [0,T]$, resulting in the optimal control $u^* = -\sigma^\top \nabla V$. Plugging this into the general HJB equation (53), we readily get the PDE in (47). □

OCDBGM (cont.)

Theorem A.5 (Verification theorem). *Let V be a solution to the HJB equation in (47) with terminal condition $V(\cdot, T) = g$. Further, let $t \in [0,T]$ and define the set of admissible controls by*

$$\mathcal{U} := \left\{ u \in C^1(\mathbb{R}^d \times [t,T], \mathbb{R}^d) \colon \sup_{(x,s)\in\mathbb{R}^d\times[t,T]} \frac{\|u(x,s)\|}{1+\|x\|} < \infty \right\}. \tag{58}$$

For every control $u \in \mathcal{U}$ let $Z^u = (Z^u_s)_{s\in[t,T]}$ be the solution to the controlled SDE

$$\mathrm{d}Z^u_s = (\sigma u + b)\,(Z^u_s, s)\,\mathrm{d}s + \sigma(s)\,\mathrm{d}B_s \tag{59}$$

and let the cost of the control u be defined by

$$J(u;t) := \mathbb{E}\left[\int_t^T \left(h + \frac{1}{2}\|u\|^2 \right)(Z^u_s, s)\,\mathrm{d}s + g(Z^u_T) \middle| Z^u_t \right]. \tag{60}$$

Then for every $u \in \mathcal{U}$ it holds almost surely that

$$V(Z^u_t, t) + \mathbb{E}\left[\frac{1}{2} \int_t^T \left\|\sigma^\top \nabla V + u\right\|^2 (Z^u_s, s)\,\mathrm{d}s \middle| Z^u_t \right] = J(u;t). \tag{61}$$

In particular, this implies that $V(Z^u_t, t) = \min_{u\in\mathcal{U}} J(u;t)$ almost surely, where the unique minimum is attained by $u^* := -\sigma^\top \nabla V$.

OCDBGM (cont.)

Theorem A.7 (Girsanov theorem). *For every control $u \in \mathcal{U}$ let $Y^u = (Y^u_s)_{s\in[0,T]}$ be the solution to the controlled SDE*

$$\mathrm{d}Y^u_s = (\sigma u - \mu)(Y^u_s, s)\,\mathrm{d}s + \sigma(s)\,\mathrm{d}B_s. \tag{68}$$

For $u, v \in \mathcal{U}$, it then holds that

$$\log \frac{\mathrm{d}\mathbb{P}_{Y^u}}{\mathrm{d}\mathbb{P}_{Y^v}}(Y^u) = \mathcal{R}_0^{u-v}(Y^u) + \int_0^T (u-v)(Y^u_s, s)\cdot \mathrm{d}B_s \tag{69}$$

and, in particular, that

$$D_{\mathrm{KL}}(\mathbb{P}_{Y^u}|\mathbb{P}_{Y^v}) = \mathbb{E}\left[\log \frac{\mathrm{d}\mathbb{P}_{Y^u}}{\mathrm{d}\mathbb{P}_{Y^v}}(Y^u)\right] = \mathbb{E}\left[\mathcal{R}_0^{u-v}(Y^u)\right]. \tag{70}$$

Lemma A.8 (KL divergence and disintegration). *Let X, Y, Z be diffusion processes. Further, let Z be such that $Z_0 \sim Y_0$ and*[11] $\mathrm{d}Z = \mathrm{d}X$. *Then it holds that*

$$D_{\mathrm{KL}}(\mathbb{P}_Y|\mathbb{P}_X) = D_{\mathrm{KL}}(\mathbb{P}_Y|\mathbb{P}_Z) + D_{\mathrm{KL}}(\mathbb{P}_{Y_0}|\mathbb{P}_{X_0}). \tag{71}$$

Proof. Let $\mathbb{P}_{Y^x}$ be the path space measure of the process Y with initial condition $Y_0 = x \in \mathbb{R}^d$, let $\mathbb{P}_{Y_0}$ be the marginal at time $t = 0$, and similarly define the corresponding quantities for the processes X and Z. Since our assumptions guarantee that $\mathbb{P}_{X^x} = \mathbb{P}_{Z^x}$ and $\mathbb{P}_{Z_0} = \mathbb{P}_{Y_0}$, the disintegration theorem, see, e.g., Léonard (2014), shows that

$$\frac{\mathrm{d}\mathbb{P}_Y}{\mathrm{d}\mathbb{P}_X} = \frac{\mathrm{d}\mathbb{P}_{Y^x}}{\mathrm{d}\mathbb{P}_{X^x}}\frac{\mathrm{d}\mathbb{P}_{Y_0}}{\mathrm{d}\mathbb{P}_{X_0}} = \frac{\mathrm{d}\mathbb{P}_{Y^x}}{\mathrm{d}\mathbb{P}_{Z^x}}\frac{\mathrm{d}\mathbb{P}_{Y_0}}{\mathrm{d}\mathbb{P}_{Z_0}}\frac{\mathrm{d}\mathbb{P}_{Y_0}}{\mathrm{d}\mathbb{P}_{X_0}} = \frac{\mathrm{d}\mathbb{P}_Y}{\mathrm{d}\mathbb{P}_Z}\frac{\mathrm{d}\mathbb{P}_{Y_0}}{\mathrm{d}\mathbb{P}_{X_0}}, \tag{72}$$

which implies the claim. □

OCDBGM (cont.)

Proposition A.9 (Optimal path space measure). *We define the work functional $\mathcal{W}\colon C([0,T],\mathbb{R}^d) \to \mathbb{R}$ for all suitable stochastic processes Y by*

$$\mathcal{W}(Y) := \mathcal{R}_\mu^0(Y) - \log \frac{p_{X_0}(Y_T)}{p_{X_T}(Y_0)}, \tag{73}$$

where $\mathcal{R}_\mu^0(Y^0)$ is as in (9) with $u = 0$. Further, let the path space measure $\mathbb{Q}$ be defined via the Radon-Nikodym derivative

$$\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}_{Y^0}} = \exp\left(-\mathcal{W}\right). \tag{74}$$

Then it holds that $\mathbb{Q} = \mathbb{P}_{Y^{u^*}}$, where $u^* := \sigma^\top \nabla \log \bar{p}_X$ as in Corollary 2.5, and for every $u \in \mathcal{U}$ we have that

$$\log \frac{\mathrm{d}\mathbb{P}_{Y^u}}{\mathrm{d}\mathbb{P}_{Y^{u^*}}}(Y^u) = \mathcal{R}_\mu^u(Y^u) + \int_0^T u(Y_s^u) \cdot \mathrm{d}B_s + \log \frac{p_{X_T}(Y_0^u)}{p_{X_0}(Y_T^u)}. \tag{75}$$

In particular, the expected variational gap

$$G(u) := \mathbb{E}\left[\log p_{X_T}(Y_0^u)\right] - \mathbb{E}\left[\log p_{X_0}(Y_T^u) - \mathcal{R}_\mu^u(Y^u)\right] \tag{76}$$

of the ELBO in Corollary 2.5 satisfies that

$$G(u) = D_{\mathrm{KL}}(\mathbb{P}_{Y^u}|\mathbb{P}_{Y^{u^*}}) = D_{\mathrm{KL}}(\mathbb{P}_{Y^u}|\mathbb{P}_{\bar{X}}) - D_{\mathrm{KL}}(\mathbb{P}_{Y_0^u}|\mathbb{P}_{X_T}). \tag{77}$$

Proof. Similar to computations in Nüsken & Richter (2021), we may compute

$$\log \frac{\mathrm{d}\mathbb{P}_{Y^u}}{\mathrm{d}\mathbb{Q}}(Y^u) = \log\left(\frac{\mathrm{d}\mathbb{P}_{Y^u}}{\mathrm{d}\mathbb{P}_{Y^0}}\frac{\mathrm{d}\mathbb{P}_{Y^0}}{\mathrm{d}\mathbb{Q}}\right)(Y^u) \tag{78a}$$

$$= \mathcal{R}_0^u(Y^u) + \int_0^T u(Y_s^u) \cdot \mathrm{d}B_s + \mathcal{R}_\mu^0(Y^u) - \log \frac{p_{X_0}(Y_T^u)}{p_{X_T}(Y_0^u)} \tag{78b}$$

$$= \mathcal{R}_\mu^u(Y^u) + \int_0^T u(Y_s^u) \cdot \mathrm{d}B_s + \log \frac{p_{X_T}(Y_0^u)}{p_{X_0}(Y_T^u)}, \tag{78c}$$

where we used the Girsanov theorem (Theorem A.7) and (74) in (78b). This implies that

$$D_{\mathrm{KL}}(\mathbb{P}_{Y^u}|\mathbb{Q}) := \mathbb{E}\left[\log \frac{\mathrm{d}\mathbb{P}_{Y^u}}{\mathrm{d}\mathbb{Q}}(Y^u)\right] = \mathbb{E}\left[\mathcal{R}_\mu^u(Y^u) - \log p_{X_0}(Y_T^u)\right] + \mathbb{E}\left[\log p_{X_T}(Y_0^u)\right]. \tag{79}$$

Comparing to Theorem 2.4, we realize that the above KL divergence is equivalent to the control costs up to $\mathbb{E}[-V(Y_0^u, 0)] = \mathbb{E}[\log p_{X_T}(Y_0^u)]$. Using (8), we thus conclude that

$$D_{\mathrm{KL}}(\mathbb{P}_{Y^{u^*}}|\mathbb{Q}) = 0 \tag{80}$$

OCDBGM (cont.)

Proposition A.10 (Forward KL divergence). *Let us consider the inference SDE and the controlled generative SDE as in (14), i.e.,*

$$dY_s = f(Y_s, s)\,ds + \sigma(s)\,dB_s, \quad Y_0 \sim \mathcal{D}, \quad dX_s^u = (\bar{\sigma}\bar{u} - \bar{f})(X_s^u, s)\,ds + \bar{\sigma}(s)\,dB_s, \tag{81}$$

and the associated path space measures $\mathbb{P}_Y$ and $\mathbb{P}_{\bar{X}^u}$. Then the expected variational gap is given by[12]

$$\begin{aligned} G(u) &:= \mathbb{E}\left[\log p_{X_T^u}(Y_0)\right] - \mathbb{E}\left[\log p_{X_0^u}(Y_T) - \mathcal{R}^u_{\sigma u - f}(Y)\right] && \text{(82a)} \\ &= D_{\mathrm{KL}}(\mathbb{P}_Y|\mathbb{P}_{\bar{X}^u}) - D_{\mathrm{KL}}(\mathbb{P}_{Y_0}|\mathbb{P}_{X_T^u}). && \text{(82b)} \end{aligned}$$

Proof. Let us consider the SDE

$$dY_s^{u,\mu} = (\sigma u - \mu)(Y_s^{u,\mu}, s)\,ds + \sigma(s)\,dB_s, \quad Y_0^{u,\mu} \sim \mathcal{D}. \tag{83}$$

With the choice $\mu = \sigma u - f$, we then have $Y^{u,\sigma u - f} = Y$, which actually does not depend on u anymore, and $dY^{u^*,\sigma u - f} = d\bar{X}^u$, where $u^* = \sigma^\top \nabla \log \bar{p}_{X^u}$. Together with the fact that $Y_0 \sim Y_0^{u^*,\sigma u - f}$, Lemma A.8 implies that

$$D_{\mathrm{KL}}(\mathbb{P}_{Y^{u,\sigma u - f}}|\mathbb{P}_{Y^{u^*,\sigma u - f}}) = D_{\mathrm{KL}}(\mathbb{P}_Y|\mathbb{P}_{\bar{X}^u}) - D_{\mathrm{KL}}(\mathbb{P}_{Y_0}|\mathbb{P}_{X_T^u}) \tag{84}$$

and Proposition A.9 (with $\mu = \sigma u - f$) yields the desired expression. $\square$

OCDBGM (cont.)

A.9 ELBO formulations

Here we provide details on the connection of the ELBO to the denoising score matching objective. Using the reparametrization in (14), i.e. taking $\mu := \sigma u - f$, Corollary 2.5 yields that

$$\log p_{X_T^u}(Y_0) \geq \mathbb{E}\left[\log p_{X_0^u}(Y_T) - \mathcal{R}^u_{\sigma u - f}(Y) \middle| Y_0\right]. \tag{85}$$

The next lemma shows that the expected negative ELBO in (85) equals the denoising score matching objective (Vincent, 2011) up to a constant, which does not depend on the control u.

Lemma A.11 (Connection to denoising score matching). *Let us define the denoising score matching objective by*

$$\mathcal{L}_{\mathrm{DSM}}(u) := \frac{T}{2}\mathbb{E}\left[\left\|u(Y_\tau, \tau) - \sigma^\top(\tau)\nabla \log p_{Y_\tau|Y_0}(Y_\tau|Y_0)\right\|^2\right]. \tag{86}$$

Then it holds that

$$\mathcal{L}_{\mathrm{DSM}}(u) = -\mathbb{E}\left[\log p_{X_0^u}(Y_T) - \mathcal{R}^u_{\sigma u - f}(Y)\right] + C, \tag{87}$$

with

$$C := \mathbb{E}\left[\log p_{X_0^u}(Y_T) + T\,\mathrm{div}(f)(Y_\tau, \tau) + \frac{T}{2}\left\|\sigma^\top(\tau)\nabla \log p_{Y_\tau|Y_0}(Y_\tau|Y_0)\right\|^2\right], \tag{88}$$

where $\tau \sim \mathcal{U}([0, T])$ and $p_{Y_\tau|Y_0}$ denotes the conditional density of Y_τ given Y_0.

OCDBGM (cont.)

Proof. The proof closely follows the one in Huang et al. (2021, Appendix A). For notational convenience, let us first define the abbreviations

$$p(x,s) := p_{Y_s|Y_0}(x|Y_0) \quad \text{and} \quad r(x,s) := u(x,s)\cdot\left(\sigma^\top(s)\nabla\log p(x,s)\right) \tag{89}$$

for every $x \in \mathbb{R}^d$ and $s \in [0,T]$. Note that we have

$$\frac{1}{2}\left\|u - \sigma^\top\nabla\log p\right\|^2 = \frac{1}{2}\left\|u\right\|^2 - r + \frac{1}{2}\left\|\sigma^\top\nabla\log p\right\|^2, \tag{90}$$

which shows that

$$\mathcal{L}_{\mathrm{DSM}}(u) = \mathbb{E}\left[T\left(\frac{1}{2}\left\|u\right\|^2 - r\right)(Y_\tau,\tau)\right] + \mathbb{E}\left[\frac{T}{2}\left\|\sigma^\top(\tau)\nabla\log p(Y_\tau,\tau)\right\|^2\right]. \tag{91}$$

Focusing on the first term on the right-hand side, Fubini's theorem and a Monte Carlo approximation establish that, under mild regularity conditions, it holds that

$$\mathbb{E}\left[T\left(\frac{1}{2}\left\|u\right\|^2 - r\right)(Y_\tau,\tau)\right] = \mathbb{E}\left[\mathcal{R}^u_{-f}(Y) + T\operatorname{div}(f)(Y_\tau,\tau)\right] - \int_0^T \mathbb{E}[r(Y_s,s)]\,\mathrm{d}s, \tag{92}$$

where the expectation on the left-hand side is over the random variable $(Y_\tau,\tau) \sim p_{Y_\tau|\tau}p_\tau$, where p_τ is the density of $\tau \sim \mathcal{U}([0,T])$ and $p_{Y_\tau|\tau}$ is the conditional density of Y_τ given τ.

Together with (91), this implies that

$$\mathcal{L}_{\mathrm{DSM}}(u) = -\mathbb{E}\left[\log p_{X_0^u}(Y_T) - \mathcal{R}^u_{-f}(Y)\right] - \int_0^T \mathbb{E}[r(Y_s,s)]\,\mathrm{d}s + C. \tag{93}$$

Focusing on the term $r(Y_s,s)$ for fixed $s \in [0,T]$, it remains to show that

$$\mathbb{E}[r(Y_s,s)] = -\mathbb{E}\left[\operatorname{div}(\sigma u)(Y_s,s)\right]. \tag{94}$$

Using the identities for divergences in Section A.2, one can show that

$$\operatorname{div}(\sigma u p) - \operatorname{div}(\sigma u)p = (\sigma u)\cdot\nabla p = (\sigma u)\cdot(p\,\nabla\log p) = rp. \tag{95}$$

OCDBGM (cont.)

Further, Stokes' theorem guarantees that under suitable assumptions it holds that

$$\int_{\mathbb{R}^d} \operatorname{div}(\sigma u p)(x,s)\, \mathrm{d}x = 0. \tag{96}$$

Thus, using (95) and (96), we have that

$$-\mathbb{E}\left[\operatorname{div}(\sigma u)(Y_s,s)|Y_0\right] = -\int_{\mathbb{R}^d} \operatorname{div}(\sigma u)(x,s) p(x,s)\, \mathrm{d}x \tag{97a}$$

$$= \int_{\mathbb{R}^d} r(x,s)p(x,s)\mathrm{d}x = \mathbb{E}\left[r(Y_s,s)\middle|Y_0\right]. \tag{97b}$$

Combining this with (93) finishes the proof. □

OCDBGM (cont.)

Corollary A.12 (Reverse KL divergence). *Let X^u and Y be defined by (17). Then, for every $u \in \mathcal{U}$ it holds that*

$$\log \frac{\mathrm{d}\mathbb{P}_{X^u}}{\mathrm{d}\mathbb{P}_{X^{u^*}}}(X^u) = \mathcal{R}^{\bar{u}}_{\bar{f}}(X^u) + \int_0^T \bar{u}(X^u_s)\cdot \mathrm{d}B_s + \log \frac{p_{Y_T}(X^u_0)}{\rho(X^u_T)} + \log \mathcal{Z}, \tag{98}$$

where $u^* := \sigma^\top \nabla \log p_Y$. In particular, it holds that

$$D_{\mathrm{KL}}(\mathbb{P}_{X^u}|\mathbb{P}_{\bar{Y}}) = D_{\mathrm{KL}}(\mathbb{P}_{X^u}|\mathbb{P}_{X^{u^*}}) + D_{\mathrm{KL}}(\mathbb{P}_{X^u_0}|\mathbb{P}_{Y_T}) = \mathcal{L}_{\mathrm{DIS}}(u) + \log \mathcal{Z}, \tag{99}$$

where

$$\mathcal{L}_{\mathrm{DIS}}(u) := \mathbb{E}\left[\mathcal{R}^{\bar{u}}_{\bar{f}}(X^u) + \log \frac{p_{X^u_0}(X^u_0)}{\rho(X^u_T)}\right]. \tag{100}$$

This further implies that

$$D_{\mathrm{KL}}(\mathbb{P}_{X^u_0}|\mathbb{P}_{Y_T}) - \log \mathcal{Z} = \min_{u\in\mathcal{U}} \mathcal{L}_{\mathrm{DIS}}(u), \tag{101}$$

and, assuming that $X^u_0 \sim Y_T$, the minimizing control u^* guarantees that $X^{u^*}_T \sim \mathcal{D}$.

Proof. Analogously to the proof of Lemma 2.3 and noting the interchanged roles of X^u and Y, we see that $\log \bar{p}_Y$ satisfies the HJB equation

$$\partial_t p = -\operatorname{Tr}\left(\bar{D}\nabla^2 p\right) + \bar{f}\cdot\nabla p + \operatorname{div}(\bar{f})p, \quad p(\cdot,T) = \log p_{Y_0} = \frac{\rho}{\mathcal{Z}}. \tag{102}$$

We can now proceed analogously to the proofs of Theorem 2.4 and Proposition 2.6. Using the identity

$$D_{\mathrm{KL}}(\mathbb{P}_{X^u_0}|\mathbb{P}_{Y_T}) = \mathbb{E}\left[\log \frac{p_{X^u_0}(X^u_0)}{p_{Y_T}(X^u_0)}\right], \tag{103}$$

this proves the statements in (99) and (101). The last statement follows from Theorem A.1 and the fact that the minimizer $u^* := \sigma^\top\nabla \log p_Y$ is the scaled score function. □

OCDBGM (cont.)

A.10.1 DDS as a special case of DIS

In this section, we compare the *denoising diffusion sampler* (DDS) suggested in Vargas et al. (2023a) to our *time-reversed diffusion sampler* (DIS). We present a numerical comparison in Table 1 and refer to Richter & Berner (2024, Section 3) and Vargas & Nüsken (2023) for further details. In particular, these works show that DDS can be derived using a reference process with known time-reversal. In the following, we outline how DDS can be considered a special case of DIS. To this end, we note that DDS considers the process

$$\mathrm{d}X_s = \left(-\overleftarrow{\beta}(s)X_s + 2\eta^2\overleftarrow{\beta}(s)\left(\overleftarrow{\Phi}(X_s,s) + \frac{X_s}{\eta^2}\right)\right)\mathrm{d}s + \eta\sqrt{2\overleftarrow{\beta}(s)}\,\mathrm{d}B_s \tag{104a}$$

$$= \left(\overleftarrow{\beta}(s)X_s + 2\eta^2\overleftarrow{\beta}(s)\overleftarrow{\Phi}(X_s,s)\right)\mathrm{d}s + \eta\sqrt{2\overleftarrow{\beta}(s)}\,\mathrm{d}B_s, \tag{104b}$$

with $X_0 \sim \mathcal{N}(0,\eta^2\mathrm{I})$, see also (136). To ease notation, let us define $\sigma(t) := \eta\sqrt{2\beta(t)}\mathrm{I}$, $f(x,t) = -\beta(t)x$, and $u(x,t) := \sigma(t)^\top\Phi(x,t)$ to get

$$\mathrm{d}X_s^u = (-\overleftarrow{f} + \overleftarrow{\sigma}\overleftarrow{u})(X_s^u,s)\,\mathrm{d}s + \overleftarrow{\sigma}(s)\,\mathrm{d}B_s \tag{105}$$

as in (17). We added the superscript u to the process in order to indicate the dependence on the control u. For DDS, the loss

$$\mathcal{L}_{\mathrm{DDS}}(u) := \mathbb{E}\left[\int_0^T \frac{1}{2}\left\|\overleftarrow{u}(X_s^u,s) + \frac{\overleftarrow{\sigma}(s)X_s^u}{\eta^2}\right\|^2 \mathrm{d}s + \log\frac{\mathcal{N}(X_T^u;0,\eta^2\mathrm{I})}{\rho(X_T^u)}\right] \tag{106}$$

is considered. In order to show the mathematical equivalence of DDS and DIS for the special choice of σ and f, we may apply Itô's lemma to the function $V(x,t) = \frac{1}{2}\|x\|^2$. This yields that

$$\mathbb{E}\left[\frac{1}{2}\|X_T^u\|^2 - \frac{1}{2}\|X_0^u\|^2\right] = \mathbb{E}\left[\int_0^T \left(-\overleftarrow{f} + \overleftarrow{\sigma}\overleftarrow{u}\right)(X_s^u,s)\cdot X_s^u + \eta^2 d\overleftarrow{\beta}(s)\,\mathrm{d}s\right], \tag{107}$$

where X^u is defined in (105). Noting that

$$\log\mathcal{N}(x;0,\eta^2\mathrm{I}) = -\frac{1}{2\eta^2}\|x\|^2 - \frac{1}{2}\log\left(2\pi\eta^2\right) \tag{108}$$

and

$$\nabla\cdot\overleftarrow{f}(x,t) = \nabla\cdot(-\overleftarrow{\beta}(t)x) = -d\overleftarrow{\beta}(t), \tag{109}$$

OCDBGM (cont.)

we can rewrite (107) as

$$\mathbb{E}\left[\log \frac{\mathcal{N}(X_T^u; 0, \eta^2 \mathrm{I})}{\mathcal{N}(X_0^u; 0, \eta^2 \mathrm{I})}\right] = \mathbb{E}\left[\int_0^T \left(\frac{\overleftarrow{f} - \overleftarrow{\sigma}\overleftarrow{u}}{\eta^2}\right)(X_s^u, s) \cdot X_s^u + \operatorname{div}\left(\overleftarrow{f}\right)(X_s^u, s)\,\mathrm{d}s\right]. \tag{110}$$

Now, we observe that the special choice of σ and f implies that

$$\left(\frac{\overleftarrow{f} - \overleftarrow{\sigma}\overleftarrow{u}}{\eta^2}\right)(X_s^u, s) \cdot X_s^u = \frac{1}{2}\left\|\overleftarrow{u}(X_s^u, s)\right\|^2 - \frac{1}{2}\left\|\overleftarrow{u}(X_s^u, s) + \frac{\overleftarrow{\sigma}(s) X_s^u}{\eta^2}\right\|^2, \tag{111}$$

which, together with (110), shows that the DDS objective in (106) can be written as

$$\mathcal{L}_{\mathrm{DDS}}(u) = \mathbb{E}\left[\mathcal{R}_{\overleftarrow{f}}^{\overleftarrow{u}}(X^u) + \log \frac{\mathcal{N}(X_0^u; 0, \eta^2 \mathrm{I})}{\rho(X_T^u)}\right]. \tag{112}$$

Recalling that $p_{X_0^u} = \mathcal{N}(0, \eta^2 \mathrm{I})$, we obtain that $\mathcal{L}_{\mathrm{DDS}}(u) = \mathcal{L}_{\mathrm{DIS}}(u)$, where the DIS objective is given as in (19).

OCDBGM (cont.)

A.10.2 Optimal drifts of DIS and PIS

In this section, we analyze the optimal drift for DIS and PIS in the tractable case where the target $\rho = \mathcal{N}(m, \nu^2 I)$ is a multidimensional Gaussian. This reveals that the drift of DIS can exhibit preferable numerical properties in practice. We refer to Figures 14 and 15 for visualizations.

Using the VP SDE in Section A.13 with $\eta = 1$ (as in our experiments), the optimal drift for DIS is given by

$$\mathrm{drift}_{\mathrm{DIS}}(x,t) \overset{(17)}{=} \bar{\sigma}(t)\bar{u}^*(x,t) - \bar{f}(t) = \bar{\sigma}^2(t) \underbrace{\frac{e^{-\bar{\alpha}(t)}m - x}{1 + e^{-2\bar{\alpha}(t)}(\nu^2 - 1)}}_{=\nabla \log \bar{p}_Y(x,t)} + \frac{1}{2}\bar{\sigma}^2(t)x, \tag{113}$$

where α is as in (138), see (137). Note that for $\nu = 1$, we can rewrite the score as

$$\nabla \log \bar{p}_Y(x,t) = e^{-\bar{\alpha}(t)}m - x = (1 - e^{-\bar{\alpha}(t)})\nabla \log p_{X_0^u}(x) + e^{-\bar{\alpha}(t)}\nabla \log \rho(x), \tag{114}$$

which is reminiscent of our initialization given by a linear interpolation and, in fact, the same for the (admittedly exotic) choice $\sigma^2(t) = \frac{2}{T-t}$, see Section A.13.

In comparison, the optimal drift of PIS (using $f = 0$, i.e., a pinned Brownian motion scaled by $\sigma \in (0, \infty)$ as in the original paper) can be calculated via the Feynman-Kac formula (Zhang & Chen, 2022a; Tzen & Raginsky, 2019) and is given by

$$\mathrm{drift}_{\mathrm{PIS}}(x,t) = \sigma\bar{u}^*(x,t) = \frac{\sigma^2 T}{t\nu^2 + \sigma^2 T(T-t)}\left(m - \frac{Tx}{t}\right) + \frac{x}{t}, \tag{115}$$

see Vargas et al. (2023a, Appendix A.2). Due to the initial delta distribution, the drift of PIS can become unbounded for $t \to 0$, which causes instabilities in practice. Note that this is not the case for DIS.

OCDBGM (cont.)

Let us first rewrite the PDEs from Section 2.1 in a way that makes their numerical approximation feasible. To this end, we recall that the Fokker-Planck equation is a linear PDE and its time-reversal can be written as a generalized Kolmogorov backward equation. Specifically, in the setting of Section 3, it holds for the time-reversed density $\overleftarrow{p}_Y$ that

$$\partial_t \overleftarrow{p}_Y = \operatorname{div}\left(-\operatorname{div}\left(\overleftarrow{D}\overleftarrow{p}_Y\right) + \overleftarrow{f}\overleftarrow{p}_Y\right) = -\operatorname{Tr}\left(\overleftarrow{D}\nabla^2 \overleftarrow{p}_Y\right) + \overleftarrow{f}\cdot\nabla\overleftarrow{p}_Y + \operatorname{div}(\overleftarrow{f})\overleftarrow{p}_Y, \tag{116}$$

analogously to (5). We note that, due to the linearity of the PDE in (116), the scaled density $p := \mathcal{Z}\overleftarrow{p}_Y$ also satisfies a Kolmogorov backward equation given by

$$\partial_t p = -\operatorname{Tr}\left(\overleftarrow{D}\nabla^2 p\right) + \overleftarrow{f}\cdot\nabla p + \operatorname{div}(\overleftarrow{f})p, \quad p(\cdot, T) = \rho. \tag{117}$$

The corresponding HJB equation following from the Hopf–Cole transformation $V := -\log\left(\mathcal{Z}\overleftarrow{p}_Y\right)$ (as outlined in Section A.5) equals

$$\partial_t V = -\operatorname{Tr}(\overleftarrow{D}\nabla^2 V) + \overleftarrow{f}\cdot\nabla V - \operatorname{div}(\overleftarrow{f}) + \frac{1}{2}\left\|\overleftarrow{\sigma}^\top\nabla V\right\|^2, \quad V(\cdot, T) = -\log\rho, \tag{118}$$

in analogy to Lemma 2.3, however, with a modified terminal condition since we omitted the normalizing constant. Now, as ρ is known, viable strategies to learn u^* would be to solve the PDEs for p or V via deep learning.

The general idea for the approximation of PDE solutions via deep learning is to define loss functionals $\mathcal{L}$ such that

$$\mathcal{L}(p) \text{ is minimal } \iff p \text{ is a solution to (117)}, \tag{119}$$

or, respectively,

$$\mathcal{L}(V) \text{ is minimal } \iff V \text{ is a solution to (118)}. \tag{120}$$

This variational perspective then allows to parametrize the solution p or V by a neural network and to minimize suitable estimator versions of the loss functional $\mathcal{L}$ via gradient-based methods. We can then use automatic differentiation to compute

$$u^* := \sigma^\top\nabla\log p_Y = \sigma^\top\nabla\log\overleftarrow{p} = \sigma^\top\nabla\overleftarrow{V} \tag{121}$$

as the score is invariant to rescaling. This adds an alternative to directly optimizing the control costs in (19) in order to approximately learn $u \approx u^*$.

OCDBGM (cont.)

A.11.1 Physics informed neural networks

A general approach to obtain suitable loss functionals $\mathcal{L}$ for learning the PDEs, dating back to the 1990s (see, e.g., Lagaris et al. (1998)), is often referred to as *Physics informed neural networks* (PINNs) or *deep Galerkin method* (DGM). These approaches have recently become popular for approximating solutions to PDEs via a combination of Monte Carlo simulation and deep learning (Nüsken & Richter, 2023; Raissi et al., 2017; Sirignano & Spiliopoulos, 2018). The idea is to minimize the squared residual of the PDE in (118) for an approximation $\widetilde{V} \approx V$, i.e.,

$$\mathcal{L}_{\mathrm{PDE}}(\widetilde{V}) = \mathbb{E}\left[\left(\partial_t \widetilde{V} + \mathrm{Tr}\left(\bar{D}\nabla^2\widetilde{V}\right) - \bar{f}\cdot\nabla\widetilde{V} + \mathrm{div}(\bar{f}) - \frac{1}{2}\|\bar{\sigma}^\top\nabla\widetilde{V}\|^2\right)^2(\xi,\tau)\right], \tag{122}$$

as well as the squared residual of the terminal condition, i.e.,

$$\mathcal{L}_{\mathrm{boundary}}(\widetilde{V}) = \mathbb{E}\left[\left(\widetilde{V}(\xi,T) + \log\rho(\xi)\right)^2\right], \tag{123}$$

where (ξ,τ) is a suitable random variable distributed on $\mathbb{R}^d \times [0,T]$. This results in a loss

$$\mathcal{L}_{\mathrm{PINN}}(\widetilde{V}) = \mathcal{L}_{\mathrm{PDE}}(\widetilde{V}) + c\,\mathcal{L}_{\mathrm{boundary}}(\widetilde{V}), \tag{124}$$

OCDBGM (cont.)

A.11.2 BSDE-based methods and Feynman-Kac formula

To reduce the computational cost for the higher-order derivatives in the PINN objective, one can also develop loss functionals $\mathcal{L}$ for our specific PDEs by considering suitable stochastic representations. This can be done by applying Itô's lemma to the corresponding solutions. For instance, for the HJB equation in (118), we obtain

$$-\log \rho(X_T) = V(\xi,\tau) + \mathcal{R}_{-\bar{f}}^{\bar{\sigma}^\top \nabla V}(X) + \mathcal{S}_V(X), \tag{125}$$

where we abbreviate

$$\mathcal{S}_p(X) := \int_\tau^T \left(\sigma^\top \nabla p\right)(X_s, s) \cdot \mathrm{d}B_s. \tag{126}$$

In the above, the stochastic process X is given by the SDE

$$\mathrm{d}X_s = -\bar{f}(X_s, s)\,\mathrm{d}s + \bar{\sigma}\mathrm{d}B_s, \quad X_\tau \sim \xi, \tag{127}$$

where (ξ,τ) is again a suitable random variable distributed on $\mathbb{R}^d \times [0,T]$. Minimizing the squared residual in (125) for an approximation $\widetilde{V} \approx V$, we can thus define a viable loss functional by

$$\mathcal{L}_{\mathrm{BSDE}}(\widetilde{V}) := \mathbb{E}\left[\left(\widetilde{V}(\xi,\tau) + \mathcal{R}_{-\bar{f}}^{\bar{\sigma}^\top \nabla \widetilde{V}}(X) + \mathcal{S}_{\widetilde{V}}(X) + \log \rho(X_T)\right)^2\right], \tag{128}$$

see, e.g., Richter (2021); Nüsken & Richter (2021). The name of the loss functional $\mathcal{L}_{\mathrm{BSDE}}$ stems from the fact that (129) can also be seen as a backward stochastic differential equation (BSDE). A method to combine the PINN loss in A.11.1 with the BSDE-based loss in (128) is presented in Nüsken & Richter (2023), leading to the so-called *diffusion loss*, which allows to consider arbitrary SDE trajectory lengths.

One can argue in a similar fashion for the linear PDE in (117), where Itô's lemma establishes that

$$\rho(X_T) = p(\xi,\tau) + \int_\tau^T \left(\mathrm{div}(\bar{f})p\right)(X_s,s)\,\mathrm{d}s + \mathcal{S}_p(X). \tag{129}$$

However, for the linear Kolmogorov backward equation, we can also make use of the celebrated Feynman-Kac formula, i.e.,

$$p(\xi,\tau) = \mathbb{E}\left[\exp\left(-\int_\tau^T \mathrm{div}(\bar{f})(X_s,s)\,\mathrm{d}s\right)\rho(X_T)\middle|(\xi,\tau)\right] \tag{130}$$

to establish the loss

$$\mathcal{L}_{\mathrm{FK}}(\widetilde{p}) = \mathbb{E}\left[\left(\widetilde{p}(\xi,\tau) - \exp\left(-\int_\tau^T \mathrm{div}(\bar{f})(X_s,s)\,\mathrm{d}s\right)\rho(X_T)\right)^2\right], \tag{131}$$

see, e.g., Berner et al. (2020); Richter & Berner (2022); Beck et al. (2021). Since the stochastic integral $\mathcal{S}_p(X)$ has vanishing expectation conditioned on (ξ,τ), it can also be included in the FK-based loss in (131) to reduce the variance of corresponding estimators and improve the performance, see Richter & Berner (2022).

OCDBGM (cont.)

Importance sampling is a classical method for variance-reduction in Monte Carlo approximation that leads to unbiased estimates of corresponding quantities of interest (Liu & Liu, 2001). The idea is to sample from a proposal distribution and correct for the corresponding bias by a likelihood ratio between the proposal and the target distribution. While importance sampling is commonly used for densities, one can also employ it in path space, thereby reweighting continuous trajectories of a stochastic process (Hartmann et al., 2017). In our application, we can make use of this in order to correct for the bias introduced by the fact that u is only an approximation of the optimal u^*. Note that, in general, it holds for any suitable functional $\varphi : C([0,T],\mathbb{R}^d) \to \mathbb{R}$ that

$$\mathbb{E}\left[\varphi(X^{u^*})\right] = \mathbb{E}\left[\varphi(X^u)\frac{\mathrm{d}\mathbb{P}_{X^{u^*}}}{\mathrm{d}\mathbb{P}_{X^u}}(X^u)\right], \tag{132}$$

i.e., the expectation can be computed w.r.t. samples from an optimally controlled process X^{u^*}, even though the actual samples originate from the process X^u. The weights necessary for this can be calculated via

$$w^u := \frac{\mathrm{d}\mathbb{P}_{X^{u^*}}}{\mathrm{d}\mathbb{P}_{X^u}}(X^u) = \frac{1}{\mathcal{Z}}\exp\left(-\mathcal{R}_{\bar{f}}^{\bar{u}}(X^u) - \int_0^T \bar{u}(X_s^u)\cdot \mathrm{d}B_s - \log\frac{p_{Y_T}(X_0^u)}{\rho(X_T^u)}\right), \tag{133}$$

where we used the Radon-Nikodym derivative from Proposition A.9. The identity in (132) then yields an unbiased estimator, however, with potentially increased variance. In fact, the variance of importance sampling estimators can scale exponentially in the dimension d and in the deviation of u from the optimal u^*, thereby being very sensitive to the approximation quality of the control u, see, for instance, Hartmann & Richter (2021).

Note that, in practice, we do not know $\mathcal{Z}$, i.e., we need to consider the unnormalized weights

$$\widetilde{w}^u := w^u\mathcal{Z}. \tag{134}$$

We can then make use of the identity

$$\mathbb{E}\left[\varphi(X^{u^*})\right] = \frac{\mathbb{E}\left[\varphi(X^u)\widetilde{w}(X^u)\right]}{\mathbb{E}\left[\widetilde{w}(X^u)\right]}. \tag{135}$$

While for normalized weights as in (133) the variance of the corresponding importance sampling estimator vanishes at the optimum $u = u^*$, this is, in general, not the case for the unnormalized weights defined in (134).

In implementations, we introduce a bias in the importance sampling estimator by simulating the SDE X^u using a time-discretization, such as the Euler-Maruyama scheme, see Section A.13. Moreover, the quantity p_{Y_T} in (133) is typically intractable and we need to use the approximation $p_{X_0^u} \approx p_{Y_T}$ as in the loss $\mathcal{L}_{\mathrm{DIS}}$, see Corollary 3.1. We have illustrated the effect of importance sampling in Figure 8.

OCDBGM (cont.)

SDE: For the inference SDE Y we employ the variance-preserving (VP) SDE from Song et al. (2021) given by

$$\sigma(t) := \eta\sqrt{2\beta(t)}\,\mathrm{I} \quad \text{and} \quad f(x,t) := -\beta(t)x \tag{136}$$

with $\beta \in C([0,T],(0,\infty))$ and $\eta \in (0,\infty)$. Note that this constitutes an Ornstein-Uhlenbeck process with conditional density

$$p_{Y_t|Y_0}(\cdot|Y_0) = \mathcal{N}\left(e^{-\alpha(t)}Y_0, \eta^2\left(1-e^{-2\alpha(t)}\right)\mathrm{I}\right), \tag{137}$$

where

$$\alpha(t) := \int_0^t \beta(s)\mathrm{d}s. \tag{138}$$

In particular, we observe that

$$p_{Y_T} = \mathbb{E}\left[p_{Y_T|Y_0}(\cdot|Y_0)\right] \approx \mathcal{N}\left(0,\eta^2\mathrm{I}\right) \tag{139}$$

for suitable β and sufficiently large T. In practice, we choose the schedule

$$\beta(t) := \frac{1}{2}\left(\left(1-\frac{t}{T}\right)\sigma_{\min} + \frac{t}{T}\sigma_{\max}\right) \tag{140}$$

with $\eta = 1$ and sufficiently large $0 < \sigma_{\min} < \sigma_{\max}$. This motivates our choice $X_0^u \sim \mathcal{N}(0,\mathrm{I})$. We can numerically check whether (the unconditional) Y_T is indeed close to a standard normal distribution. To this end, we consider samples from $\mathcal{D}$ and let the process Y run according to the inference SDE in (3). We can now compare the empirical distributions of the different components, which each need to follow a one-dimensional Gaussian. Figure 9 shows that this is indeed the case.

Model: Similar to PIS, we employ the initial score of the inference SDE, i.e., the score of the data distribution, $\nabla \log p_{Y_0} = \nabla \log \rho$, in the parametrization of our control. Additionally, in our DIS method, we also make use of the (tractable) initial score of the generative SDE $\nabla \log p_{X_0^u} = \nabla \log \mathcal{N}(0,\mathrm{I})$. Specifically, we choose

$$u_\theta(x,t) := \Phi_\theta^{(1)}(x,t) + \Phi_\theta^{(2)}(t)\sigma(t)s(x,t), \tag{141}$$

where

$$s(x,t) := \frac{t}{T}\nabla \log p_{X_0^u}(x) + \left(1-\frac{t}{T}\right)\nabla \log \rho(x) \tag{142}$$

and $\Phi_\theta^{(1)}$ and $\Phi_\theta^{(2)}$ are neural networks with parameters θ. We use the same network architectures as PIS, which in particular uses a Fourier feature embedding (Tancik et al., 2020) for the time variable t. We initialize $\theta^{(0)}$ such that $\Phi_{\theta^{(0)}}^{(1)} \equiv 0$ and $\Phi_{\theta^{(0)}}^{(2)} \equiv 1$. The parametrization in (141) establishes that our initial control (approximately) matches the optimal control at the initial and terminal times, i.e.,

$$u_{\theta^{(0)}}(\cdot,0) = \sigma(0)^\top \nabla \log \rho = \sigma(0)^\top \nabla \log p_{Y_0} = u^*(\cdot,0) \tag{143}$$

OCDBGM (cont.)

Training: We minimize the mapping $\theta \mapsto \widehat{\mathcal{L}}_{\mathrm{DIS}}(u_\theta)$ using a variant of gradient descent, i.e., the Adam optimizer (Kingma & Ba, 2014), where $\widehat{\mathcal{L}}_{\mathrm{DIS}}$ is a Monte Carlo estimator of $\mathcal{L}_{\mathrm{DIS}}$ in (19). Given that we are now optimizing a reverse KL divergence, the expectation in $\mathcal{L}_{\mathrm{DIS}}(u_\theta)$ is over the controlled process X^{u_θ} in (17) and we need to discretize X^{u_θ} and the running costs $\mathcal{R}_{\bar{f}}^{\bar{u}_\theta}(X^{u_\theta})$ on a time grid $0 = t_0 < \cdots < t_N = T$. We use the *Euler-Maruyama* (EM) scheme given by

$$\widehat{X}_{n+1}^\theta = \widehat{X}_n^\theta + \left(\bar{\sigma}\bar{u}_\theta - \bar{f}\right)(\widehat{X}_n^\theta, t_n)\Delta t + \bar{\sigma}(t_n)\underbrace{(B_{t_{n+1}} - B_{t_n})}_{\sim\mathcal{N}(0,\Delta t\mathrm{I})}, \tag{145}$$

where $\Delta t := T/N = t_{n+1} - t_n$ is the step size. Given $\widehat{X}_0^\theta \sim X_0^{u_\theta}$, it can be shown that $\widehat{X}_N^\theta$ convergences to $X_T^{u_\theta}$ for $\Delta t \to 0$ in an appropriate sense (Kloeden & Platen, 1992). This motivates why we increase the number of steps N for our methods when the training progresses, see Table 2. A smaller step size typically leads to a better approximation but incurs increased computational costs. We also trained separate models with a fixed number of steps, but did not observe benefits justifying the additional computational effort, see Figure 10.

We compute the derivatives w.r.t. θ by automatic differentiation. In a memory-restricted setting, one could alternatively use the *stochastic adjoint sensitivity method* (Li et al., 2020; Kidger et al., 2021) to compute the gradients using adaptive SDE solvers. For a given time constraint, we did not observe better performance of stochastic adjoint sensitivity methods, higher-order (adaptive) SDE solvers, or the exponential integrator by Zhang & Chen (2022b, Section 5) over the Euler-Maruyama scheme. Our training scheme yields better results compared to the models obtained when using the default configuration in the code of Zhang & Chen (2022a), see Figure 11. For our comparisons, we thus trained models for PIS according to our scheme, see also Table 2. Note that in Section A.11 we present an alternative approach of training a control u_θ based on *physics-informed neural networks* (PINNs).

Sampling: In order to reduce the variance, we compute an exponential moving average $\bar{\theta}$ of the parameters $(\theta^{(k)})_{k=1}^K$ in Algorithm 1 during training and use it for evaluation. We evaluate our model by sampling from the prior $\widehat{X}_0^{\bar{\theta}} \sim \mathcal{N}(0, \mathrm{I})$ and simulating the generative SDE using the EM scheme, such that $\widehat{X}_N^{\bar{\theta}}$ represents an approximate sample from $\mathcal{D}$. Note that we can, in principle, choose the number of steps for the EM scheme independent from the steps used for training. We used an increasing number of steps during training, and our trained model provides good results for a range of steps, see Figures 4 and 5. Thus, we amortized the

cost of training different models for different step sizes. Finally, note that similar to Song et al. (2021), one could alternatively obtain approximate samples from $\mathcal{D}$ by simulating the *probability flow ODE*, originating from the choice $\lambda = 1$ in Theorem A.1, using suitable ODE solvers (Zhang & Chen, 2022b).

OCDBGM (cont.)

Evaluation: We evaluate the performance of our proposed method, DIS, against PIS on the following two metrics:

- *approximating normalizing constants:* We obtain a lower bound[14] on the log-normalizing constant $\log \mathcal{Z}$ by evaluating the negative control costs, i.e.,

$$\log \mathcal{Z} \geq \log \mathcal{Z} - D_{\mathrm{KL}}(\mathbb{P}_{X_0^u} | \mathbb{P}_{Y_T}) \geq -\mathcal{L}_{\mathrm{DIS}}(u_{\bar{\theta}}) \tag{146}$$

see (20). We note that, similar to the PIS method, adding the stochastic integral (which has zero expectation), i.e., considering

$$\mathcal{R}_{\bar{\mu}}^{\bar{u}}(X^u) + \log \frac{p_{X_0^u}(X_0^u)}{\rho(X_T^u)} + \int_0^T u(Y_s^u, s) \cdot \mathrm{d}B_s \tag{147}$$

yields an estimator with lower variance, see also Corollary A.12. We will compare unbiased estimates using *importance sampling* in path space, see Section A.12.

- *approximating expectations and standard deviations:* In applications, one is often interested in computing expected values of the form

$$\Gamma := \mathbb{E}\left[\gamma(Y_0)\right], \tag{148}$$

where $Y_0 \sim \mathcal{D}$ follows some distribution and $\gamma : \mathbb{R}^d \to \mathbb{R}$ specifies an observable of interest. We consider $\gamma(x) := \|x\|^2$ and $\gamma(x) := \|x\|_1 = \sum_{i=1}^d |x_i|$. We approximate (148) by creating samples $(\widehat{X}_N^{\bar{\theta},(k)})_{k=1}^K$ from our model as described in the previous paragraph, where $K \in \mathbb{N}$ specifies the sample size. We can then approximate Γ in (148) via Monte Carlo sampling by

$$\widehat{\Gamma} := \frac{1}{K} \sum_{k=1}^{K} \gamma(\widehat{X}_N^{\bar{\theta},(k)}). \tag{149}$$

Given a reference solution Γ of (148), we can now compute the relative error $|(\widehat{\Gamma} - \Gamma)/\Gamma|$, which can be viewed as an evaluation of the sample quality on a global level. Analogously, we analyze the error when approximating coordinate-wise standard deviations of the target distribution Y_0 using the samples $(\widehat{X}_N^{\bar{\theta},(k)})_{k=1}^K$.

Thank you!